

Next CW Trainer

User Manual

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Next CW Trainer – User Manual, for app version 1.3.0 (source fab962b), October 2026. Author: Christian Konecny, OE1CKO. App icon: Sia, OE1LMR.

Next CW Trainer is an independent project. Its training concept and algorithms come from the Morserino-32 firmware by Willi Kraml, OE1WKL, and the Morserino-32 team; there is no other connection to them.

This manual is also available in German.

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1 Introduction

Next CW Trainer is an Android app for learning and practicing Morse code (CW): copy training with the Koch method, sending practice with the Echo Trainer, a CW Keyer, a CW Decoder using the microphone, CW over the internet (WiFi Trx), a QSO Bot and several games. An **adaptive block flow** tracks your mistakes character by character and suggests when you are ready for a new character, more speed or shorter pauses.

Where the app comes from

Next CW Trainer is an independent hobby project by Christian Konecny, OE1CKO. Many ideas and much of the training logic come from the open-source firmware of the Morserino-32 by Willi Kraml, OE1WKL. Its training concept, with the Koch sequences, the Echo Trainer, the QSO Bot and much more, is the result of years of careful work by Willi and his team. Many thanks for that!

The logic was read out of the firmware source and rewritten for Android; the app contains no original firmware code. Beyond that there is **no connection** to Willi Kraml or the Morserino-32 team, and the app is not a product of the Morserino project.

The app was based on firmware version 9.0.0. Later firmware changes are not carried over automatically.

About this manual

This manual describes the app version printed on the title page. You can see which version you have installed under **Settings → Info** (see Info: version and build).

Settings that also exist on the Morserino have plain names in the app (for example **Character spacing** instead of "Interchar Spc"). The table Morserino terms lists which Morserino menu item belongs to which setting.

2 Getting started

2.1 Installation

The app is currently handed out as an APK file, not through the Play Store. It needs Android 8.0 or newer and a 64-bit device (practically every phone since about 2017); it cannot be installed on 32-bit-only devices.

1. Copy the APK file to your phone, for example by messenger, e-mail or USB cable.
2. Tap the file in your file manager. The first time, Android asks whether the app you install from may **install unknown apps**. That might be your file manager or browser. Allow it for that app.
3. Confirm the installation.

Updates work the same way: install the new APK over the old one, and your settings and statistics are kept. This only works if both APKs are signed with the same key. If Android reports a conflict, you first have to uninstall the old version, and that deletes your settings and statistics.

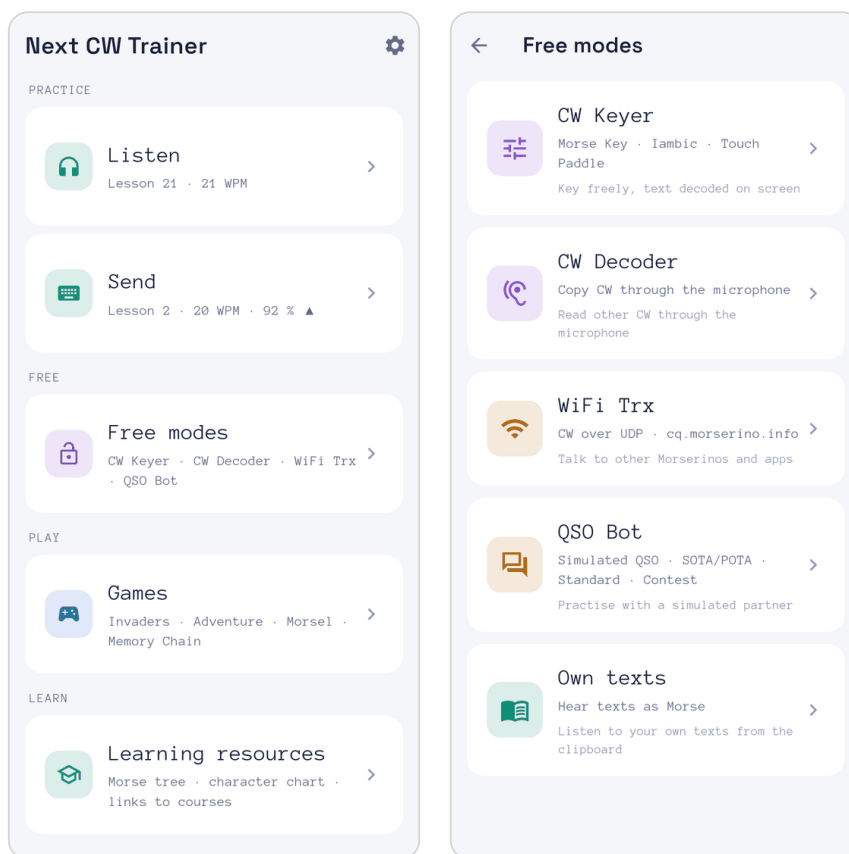
2.2 The home screen

The home screen has four groups:

Group	Tile	What for
Practice	Listen	Practice copying: CW Generator and Koch Trainer in the block flow
	Send	Practice sending: Echo Trainer. A word is played and you key it back
Free	Free modes	Opens five tiles: CW Keyer (key freely, with the text decoded on screen), CW Decoder (copy CW through the microphone), WiFi Trx (CW over the internet with other Morserinos and apps) QSO Bot (a simulated QSO partner) and Own texts (hear your own texts from the clipboard as Morse, see Own texts)
Play	Games	Morse Invaders, text adventure, Morsel, Memory Chain
Learn	Learning resources	Interactive Morse tree, character chart, links to courses and practice sites

The **Listen** and **Send** tiles show your current Koch lesson and speed. **Send** also shows the trend of your last blocks (see Trend).

The gear icon at the top right opens the **global settings** (chapter Settings). Anything that concerns only one training is set up directly inside that training.



The home screen

Free modes: five tiles

2.3 Controls inside the trainings

Each training has these icons at the top right:

- **Training settings** opens a sheet that slides up from the bottom. Every change takes effect immediately and is saved.
- **Statistics** (only in **Listen** and **Send**) shows your progress per character (see Character statistics).

While a block is running, these icons and the selection at the top are hidden to keep the screen calm. The **back button** then only ends the running block and returns you to the training's start view. A second back leaves the training.

The screen stays on while you practice.

Text size: you can **pinch** text areas that show decoded text or results to make the text larger or smaller. This works in the CW Keyer, in Send and on the result page, among others. Each area remembers its own size.

2.4 Light, dark, language

Under **Settings** → **Appearance** you choose the **Theme** (System, Light or Dark) and the app **Language** (Deutsch or English). Both take effect immediately.

The app follows the Android **font size** setting, but only up to **1.3×** the default size (step 4 of 7 on a Pixel). Larger steps look the same as that one in the app; otherwise the training screens would no longer fit. The on-screen keyboard for typing along and the Koch character row always keep their size. The Android **Display size** setting (“Make everything bigger or smaller”) has no effect in the app; it always uses the device’s default size. When a page is longer than the screen, a scrollbar at the edge shows that there is more.

2.5 Sound and volume

Set the volume with the phone’s volume keys. You set the **pitch** and the **tone softness** in the settings. When headphones, a USB audio device or a Bluetooth device is connected or disconnected, the app switches to it automatically. If you don’t want that, you can fix the output (see Audio output).

Bluetooth headphones usually add noticeable latency. That doesn’t matter for listening, but it does for sending: you hear your sidetone noticeably later than you key. For sending, use wired headphones or the speaker.

3 Basics

3.1 Speed in WPM

Speed is given in **WPM** (words per minute), based on the standard word “PARIS”. At w WPM a dit lasts exactly $1200 / w$ milliseconds, so 60 ms at 20 WPM.

3.2 Spacing in dits: character spacing and word spacing

As on the Morserino, the pauses are set **in dit lengths**:

- **Character spacing** is the pause between two characters of a word. Normal Morse is 3 dits. You can set 3 to 45.
- **Word spacing** is the pause between two words or groups. Normal Morse is 7 dits. You can set 6 to 105.

The pause between the elements (dits and dahs) *inside* a character is always 1 dit. So the characters themselves always come at the set speed, and only the pauses between them get longer. This is the **Farnsworth method**. You learn the sound of a character at full speed and still have time to think.

Next to each spacing slider the app also shows the pause **in seconds** at the current speed, for example “28 dits · 1.68 s @ 20 WPM”.

The **Listen** and **Send** trainings start with generous pauses of **28 / 40 dits**. The CW Keyer, WiFi Trx and the QSO Bot use the normal 7 dits as word spacing.

3.3 Effective speed

Because the pauses are longer, whole words arrive more slowly. The app shows this as **eff.** (effective WPM):

$\text{eff. WPM} = 50 \times \text{WPM} / (31 + 4 \times \text{character spacing} + \text{word spacing})$

Example: 20 WPM with 28/40 dits gives $50 \times 20 / (31 + 112 + 40) \approx 5$ WPM. The characters sound like 20 WPM, but you have as much time as at 5 WPM. With 3/7 dits the two values are the same.

3.4 The Koch method

With the Koch method you start with **two characters** at full speed. Once you recognise them reliably, the next character is added, then the next, until you know them all. Each step is called a **lesson**. The lesson number is the number of active characters.

The order of the characters is set by the **Koch sequence** (see Koch sequence).

When drawing random characters, the app weights them like the Morserino. Two out of three characters are drawn evenly from **all** active characters. Every third one is drawn only from the **last third**, where the most recently learned characters are. New characters therefore come up more often, without the old ones disappearing.

3.5 Prosigns

Prosigns (procedural signals) are sent as run-together letters. The app writes them in angle brackets:

Prosign	Meaning
<ka>	Start of message
<ar>	End of message (+)
<kn>	Only the station called should answer
<sk>	End of contact
<as>	Wait
<ve>	Understood
<bk>	Break, over to the other station
<err>	Error: eight dits (.)

In Listen (block flow), prosigns from **Group characters** are shown and graded as two separate letters: <ka> appears as “K A”. When copying on the on-screen keyboard you type them as two letters as well.

4 Listen: practice copying

Listen is the app's CW Generator and Koch Trainer. The app plays a **block** of groups or words, and you copy it, in one of two ways:

- **Paper:** you copy on paper. Nothing is shown on screen while it plays. Afterwards the app reveals the text and you tap what you got wrong.
- **Type:** for on the go. The app plays one word after another and you type it on the app's own on-screen keyboard (see Copying on the on-screen keyboard).

Either way you get the same result with suggestions for the next block at the end, and both count towards the same Listen statistics.

4.1 Choosing the character set and content

At the top of the start view you choose two things.

The **character set** decides which characters are practiced:

Character set	Meaning
Koch lesson	Only the characters up to your current lesson
All characters	All letters, digits, punctuation and prosigns
Practice set	Only the characters you enter yourself

The **content** decides what is played:

Content	Meaning	Available with
Random	Groups of random characters	all
Words	Common English words	Koch, All
Abbreviations	Common CW abbreviations	Koch, All
Call signs	Random, realistic call signs	All
Mixed	Words, abbreviations and groups mixed	Koch, All

With the Koch lesson, only words and abbreviations made **entirely** of characters you have already learned are used. In the first lessons there are very few of them.

4.1.1 Setting the Koch lesson

The **KOCH** slider sets the lesson. Below it you see every character of the Koch sequence. The characters active in this lesson are highlighted, and their color shows the kind of character: letters, digits and punctuation/prosigns are colored differently. Characters not yet unlocked by the lesson are dimmed.

To get to know a character (this works for the dimmed ones too, so you can listen ahead to later lessons):

- **Tap** it to play it three times at the current speed. A tile with the character and its Morse code appears in the middle of the screen, and the rest of the screen is dimmed. The dits and dahs start out gray and light up exactly while they sound. Each repetition starts from gray again. After the third repetition the tile closes by itself. A tap anywhere on the screen (or the back button) stops playback at once.
- **Long press** it to open the practice page for that character: listen and, if you like, key it back (see Practicing a single character).

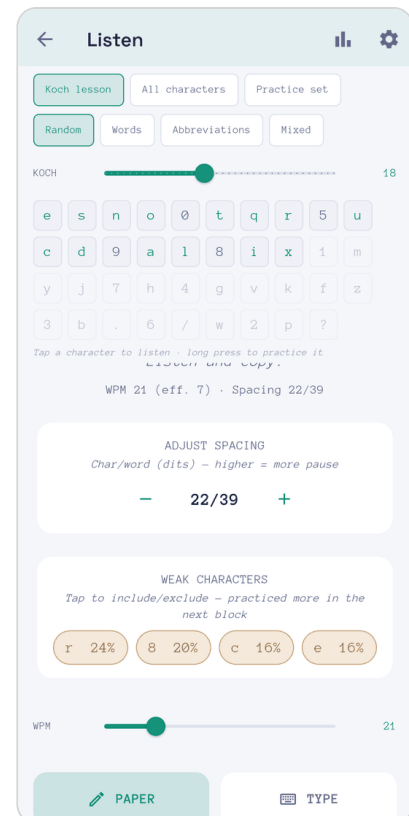
Usually you don't need to raise the lesson by hand. The block flow suggests the next character once you are ready (see When the next Koch character comes).

4.1.2 Practice set

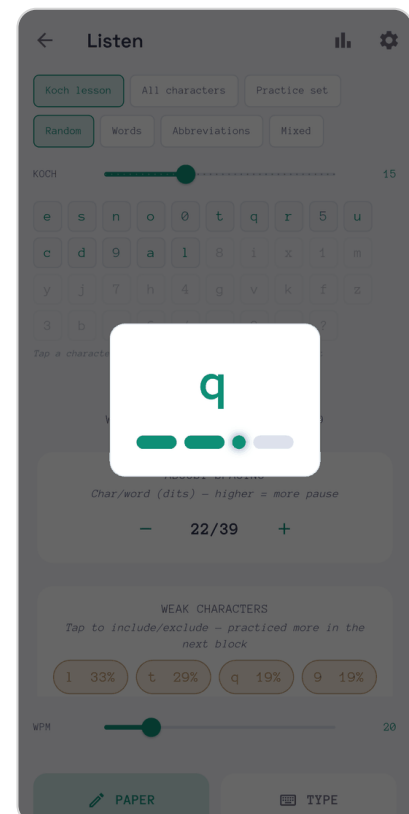
With the **Practice set** character set, an input field appears right on the start view. Enter the characters you want to practice, for example **QXZJ**. Upper/lower case and spaces don't matter, and duplicate characters are ignored. The app shows how many distinct characters it found. The practice set only offers the **Random** content.

4.2 Speed

The **WPM** slider below the practice area sets the character speed (10 to 60 WPM). You set the pauses in the * sheet under **Spacing**, or directly on the start and result pages with **Adjust spacing**.



Listen start view: character set, content, Koch lesson, weak characters, spacing and speed

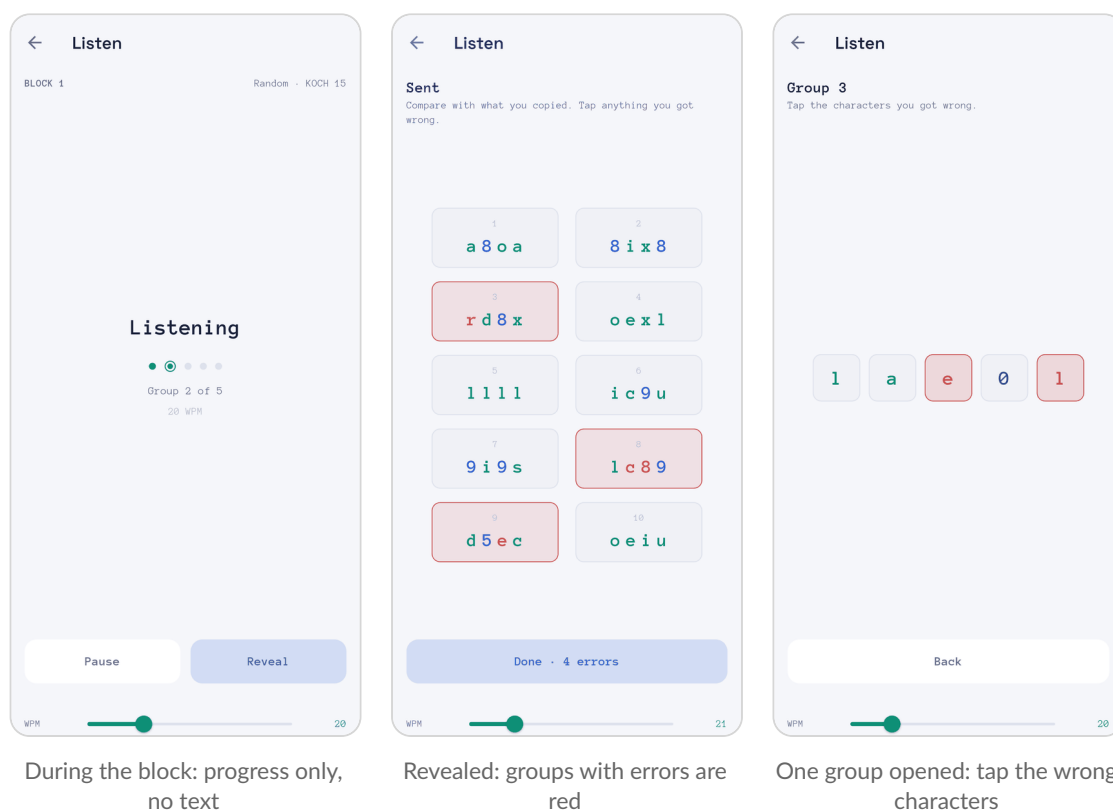


Tapping a Koch character: tile with its Morse code

4.3 How a block runs

At the bottom of the start view are two start buttons: **Paper** and **Type**. The one you used last is highlighted. This section describes **Paper**; **Type** follows further down.

1. Press **Paper**. After one second of “Get ready ...” the block begins.
2. The app plays the groups one after another, with the set word spacing in between. Only “Group *n* of *N*” (for words “Word *n* of *N*”) and the speed are shown, not the text. Copy on paper.
 - **Pause** stops after the current group, and **Resume** continues.
 - **Reveal** ends the block early and shows only the groups played so far.
3. After the last group, **Sent** appears, with each group as its own tile. Compare it with what you copied.
4. **Marking errors**: tap a group where you made a mistake. It opens large, one character at a time. Tap every character you got wrong or missed, then **Back**. Tapping a character again removes the mark. The marked characters show in red in the overview.
5. **Done · *n* errors** finishes the evaluation. The results are saved now, and the result page appears.



You set how many groups a block has in the * sheet under **Word selection** → **Groups per block** (with words, the setting is called **Words per block**). During the block and on the result page the app says **groups** for Random and **words** for all other contents.

4.3.1 Stop after each group

With **Flow** → **Stop after each group** (in the * sheet; with words it is called **Stop after each word**), the app waits after every group or word:

- **Dit** (left paddle) or the **REPEAT** button plays the same group again.
- **Dah** (right paddle) or **NEXT** plays the next group.

This matches “Stop<Next>Rep” on the Morserino. It’s handy at the start, when you want to hear a group several times. The setting only applies to **Paper**; when typing, the app waits after every word anyway.

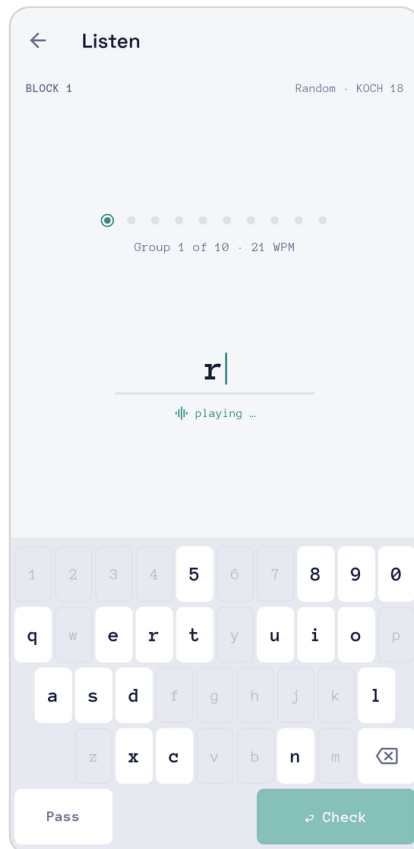
4.4 Copying on the on-screen keyboard

With **Type** you copy on the app’s own keyboard on the display instead of on paper, meant for on the go. It is not the phone’s system keyboard.

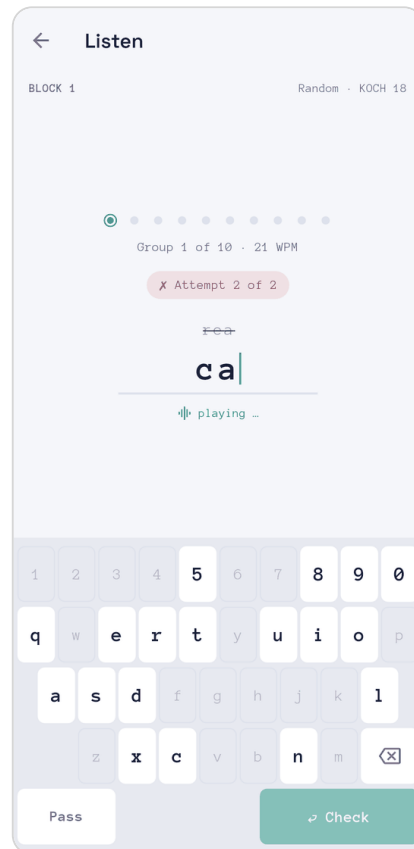
The keyboard is a QWERTY keyboard with a digit row above it. Every key is always in the same place. Only the characters the chosen character set can contain are active (bright, tappable); the others are only outlined and do nothing. If the character set contains punctuation (. , : - / = ? @ +), it gets a row of its own. While you press a key, a bubble above it shows the character large. ☒ deletes the last character. **Pass** is at the bottom left, ↵ **Check** at the bottom right. While the block runs, the WPM slider is hidden to make room for the keyboard.

How a word runs:

1. Press **Type**. After one second of “Get ready ...” the app plays the first group (or word). “playing ...” is shown below the answer line.
2. You can type along while it plays or wait until it has finished; both work. There is no time limit.
3. **Checking**: as soon as you have typed as many characters as the word has, the app checks automatically (after the word has ended plus a short 0.4 s, during which you can still correct with ☒). ↵ **Check** hands in earlier, for example when you missed a character. If you press ↵ while the word is still playing, the app checks right after the word ends.
4. ✓ **Correct**: the next word comes after about one second. Whatever you type during that second counts for the next word.
5. ✗ **Wrong**: the word is played again at once, with an empty field. Your previous attempt is shown small and struck through above it, without a hint where the error was. “Attempt *n* of *max*” shows which attempt this is.
6. **Pass**: at any time, also while the word plays. After the last wrong attempt or a pass, the app shows the solution for 2 seconds (the characters wrong in the first attempt in red) with your attempts below it, then moves on.



Typing: the keyboard, only the lesson's characters are active



Wrong: the word plays again, attempt 2 of 2

The confirmation tone (high = correct, low = wrong) follows the **Confirmation tone** setting in the * sheet of **Send**. How many attempts you get per word and whether the keys vibrate is set in the * sheet under **Flow** (see below).

After the block, **Sent** appears as with Paper. The errors of your **first** attempt are already marked red, and under each group is what you typed ("— passed" for a pass). You can tap a group and change marks, for example when you only mistyped. **Done · n errors** saves and shows the usual result page.

How it is counted:

- Only the **first** attempt of each word counts; the second is easier because you hear the word a second time.
- The app compares character by character, but not rigidly position by position: if you left out a character, only that character is wrong, not every one after it. Example: played tq~~r~~5u, typed tq5u → only **r** is wrong. A character heard wrong is wrong (cd9a~~l~~ as cb9a~~l~~ → **d** wrong). An extra typed character does not make any played character wrong.
- **Pass** in the first attempt means every character of that word is wrong, like a blank on paper.
- When typing, the pauses between words don't matter. The app therefore only suggests changes to the **character spacing** (see Pauses and speed).

The arrow at the top left cancels the block; nothing is saved then. The keyboard is portrait-only for now.



Sent after a typed block: first-attempt errors marked, your input below

4.5 The result page

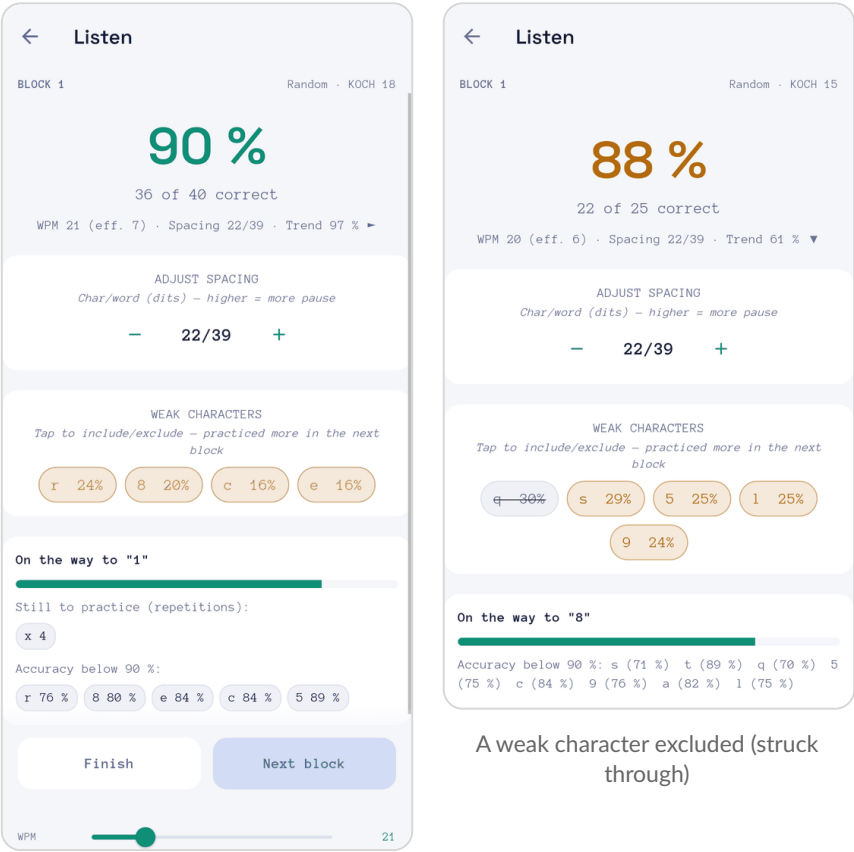
From top to bottom, the result page shows:

- **Accuracy** in percent, meaning the share of characters copied correctly, plus “x of y correct”. It is green from 90 %, yellow from 70 % and red below that.
- The **status line**: speed, effective speed, spacing and, from the sixth block on, the trend. These values already apply to the **next** block, including the ticked suggestions.
- **Adjust spacing**: – and + change character spacing and word spacing together by 1 dit each. This takes effect immediately and doesn't depend on the suggestions.
- **Weak characters**: the characters that cause you the most errors over time, with their error rate (see Weak characters).
- **On the way to “X”** (Koch lesson only): what you still need before the next character (see Progress card).
- **Suggestions**: what the app recommends for the next block (see Accepting, rejecting, adjusting suggestions).

At the bottom there are two buttons:

- **Next block** applies the ticked suggestions and starts the next block right away.
- **Finish** also applies the ticked suggestions but doesn't start a new block.

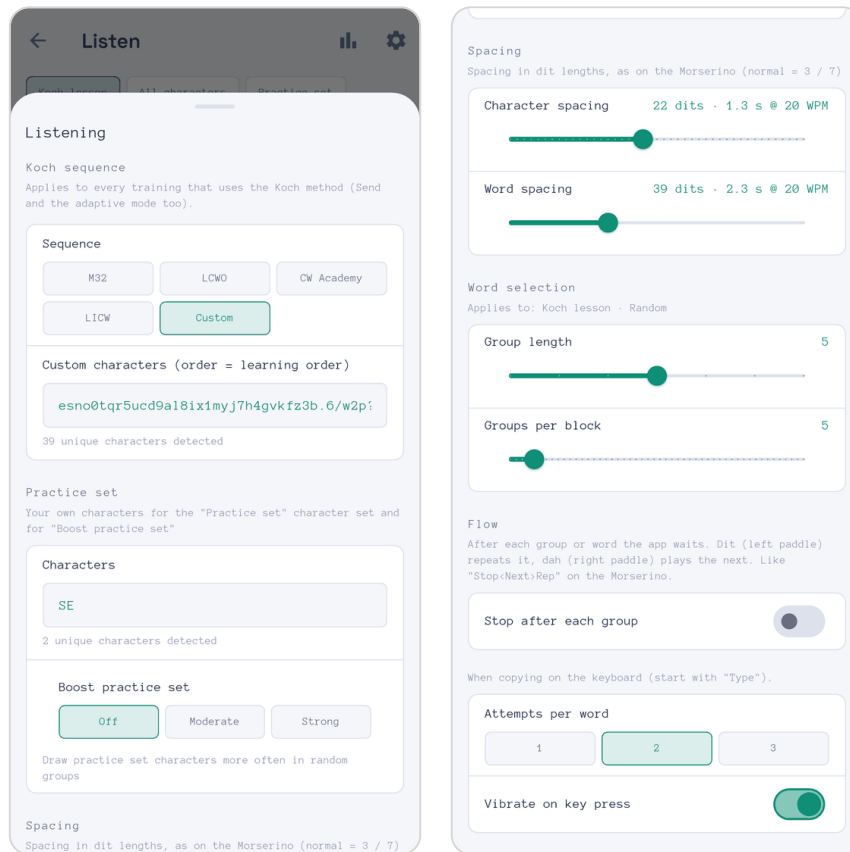
How the app arrives at its suggestions is explained in the chapter Adaptive mode.



Result page with accuracy, spacing, weak characters and progress card

4.6 Settings in the ⚙ sheet (Listen)

The ⚙ sheet of **Listen** has the following sections. Values in **bold** are the defaults.



* sheet: Koch sequence and practice set

* sheet: Spacing, Word selection, Flow

4.6.1 Koch sequence

Only shown when the **Koch lesson** character set is selected. This setting applies to **all** trainings and games that use the Koch method. For a description see Koch sequence.

4.6.2 Practice set

Setting	Meaning	Values
Characters	The practice set characters. This is the same field as on the start view with the Practice set character set	any characters
Boost practice set	Draws practice set characters more often in random groups (see below)	Off / Moderate / Strong

This is how **Boost practice set** works. For each character of a random group, the app draws up to 3 times (Moderate) or 8 times (Strong) until it gets a character from the practice set. If that doesn't happen, the last character drawn stays. The practice set characters come up more often, but the others don't disappear.

Boost practice set works in **Listen** (Koch lesson or All characters · Random) and in **Send** (All characters · Random). In **Listen** the practice set is combined with the weak characters: the characters from both lists are boosted, at the higher of the two levels. The more characters come together, the less each single one stands out.

4.6.3 Spacing

Setting	Meaning	Values
Character spacing	Pause between characters, in dits	3–45 (28)
Word spacing	Pause between groups/words, in dits	6–105 (40)

Word spacing can never be smaller than character spacing. If you move character spacing past it, word spacing is pulled along.

4.6.4 Word selection

Only the settings that fit the chosen content are shown. The line “Applies to: ...” tells you which combination you are setting up.

Setting	Meaning	Values
Group characters	Only with All characters • Random : which character classes are drawn from	All / Letters / Digits / Punctuation / Prosigs / Letters+Digits / Digits+Punct. / Punct.+Prosigs / Letters+Digits+Punct. / Digits+Punct.+Prosigs
Group length	Characters per random group (only with Random)	2–8 (5)
Max word length	Only words up to this length (with Words and Mixed)	all , 1–8
Max abbreviation length	Only abbreviations up to this length (with Abbreviations and Mixed)	all , 2–6
Groups per block / Words per block	Number of groups (with Random) or words in a block	1–50 (10)

4.6.5 Flow

Setting	Meaning	Values
Stop after each group / Stop after each word	Paper only: wait after each one, dit = repeat, dah = next	Off / On
Attempts per word	Type only: how often you may try a word; 1 = no second attempt	1 / 2 / 3

Setting	Meaning	Values
Vibrate on key press	Type only: short vibration on every active key	Off / On

4.6.6 Adaptive mode

These are the thresholds the app bases its suggestions on. They apply to **Listen and Send** together and are described in detail in Adaptive mode settings.

The screenshot displays the 'Adaptive mode' settings interface. At the top, there is a slider for 'Groups per block' set to 5. Below this is a section titled 'Flow' with explanatory text: 'After each group or word the app waits. Dit (left paddle) repeats it, dah (right paddle) plays the next. Like "Stop:Next:Rep" on the Morserino.' A toggle switch for 'Stop after each group' is currently turned off. The next section, 'When copying on the keyboard (start with "Type")', features a 'Attempts per word' selector with buttons for 1, 2 (selected), and 3. Below this is a 'Vibrate on key press' toggle switch, which is turned on. The final section, 'Adaptive mode', includes a description: 'Thresholds controlling how the adaptive mode adjusts speed and spacing and when the next Koch character unlocks.' It contains three sliders: 'Success threshold low/high' set between 70% and 90%, 'EMA smoothing' set at 30%, and 'Occurrences for unlock' set at 20.

* sheet: Adaptive mode

5 Send: Echo Trainer

Send is the Echo Trainer. The app plays a word or group, and you key it back with the paddle. That can be the touch paddle or a real paddle (see Paddle and Morse key). If you get it right, the next word comes. If not, it is repeated.

Send has its **own profile**, independent of Listen. It keeps its own Koch lesson, speed, spacing, practice set and character statistics. What you mix up when listening isn't necessarily what you get wrong when sending.

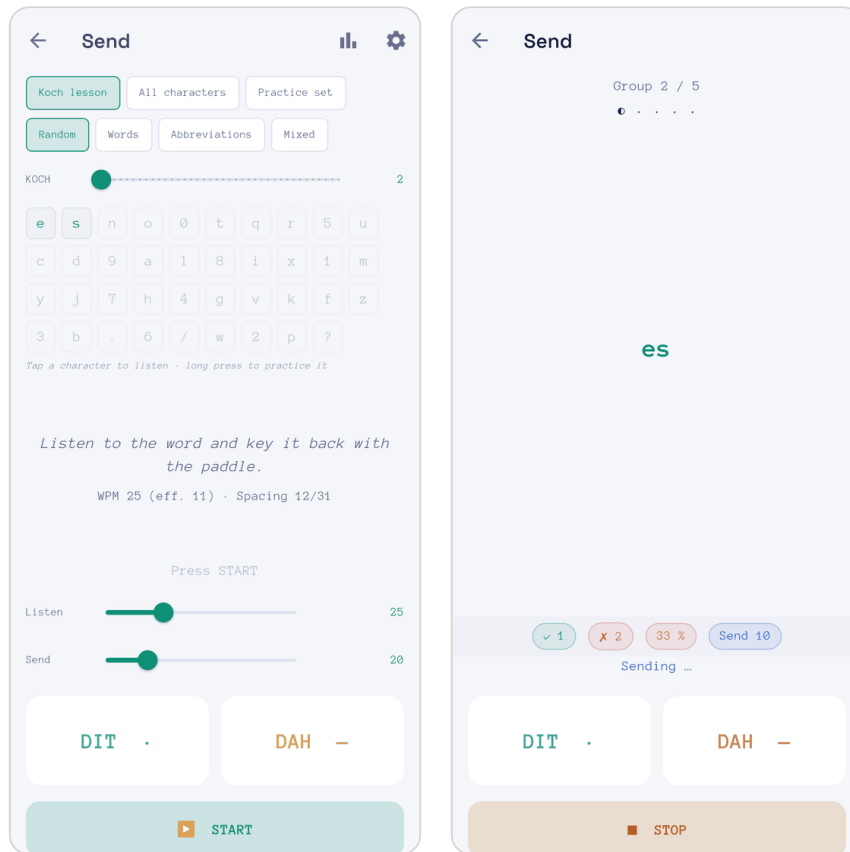
5.1 Start view

At the top you choose the **character set** and **content** as in Listen (see Choosing the character set and content). With the Koch lesson you also choose the lesson. Tapping a character works as in Listen.

Below that, two sliders set the speeds:

- **Listen** is the speed at which the word is played to you (10 to 60 WPM).
- **Send** is the highest speed at which your answer is expected (see Sending speed). "same as prompt" means the same speed.

At the bottom are the paddles or the key, and **Start**.



Send start view

While answering: prompt (Prompt = Both), attempt and speed

5.2 How a word runs

1. After **Start** the app waits 2 seconds, then plays the first word.
2. **Your answer:** key the word back. Your sidetone is shifted by half a tone so you can tell prompt and answer apart. This is the **Tone shift** setting.
3. **When you have to start:** from the end of the prompt you have about 1.4 seconds, plus one character pause, plus a third of the word pause, plus the **Think time** (default 8 s) to **begin** your answer. If nothing comes by then, the word counts as wrong.
4. **When the answer ends:** once you have started, the think time no longer applies. The answer is complete as soon as you leave a 7-dit word pause, at sending speed.
5. **Correcting:** key <err> (eight dits) or four e in a row to clear your answer so far and start again. The exception is when the word itself continues with another e at that point. Then the e counts as a normal character.
6. **Grading:**
 - **✓ Correct**, optionally with a confirmation tone. The next word comes after a little more than a second.
 - **✗ Wrong:** the word is played again. "Attempt *n* of *max*" shows which attempt this is. Once all **Repeats** are used up, the app shows the correct word for two seconds and moves on.

Whether the prompt is played, shown or both is set with **Prompt** (see below).

5.3 Sending speed

On the Morserino this setting is called “Echo Speed Max”. It limits the speed at which **your answer** is expected. The prompt still plays at the listening speed.

Example: Listen at 25 WPM and Send at 18 WPM means you hear fast but may answer more slowly. The sending speed sets how fast the keyer produces your dits and dahs and how long a word pause has to be. At the far left (“same as prompt”), the listening speed applies to the answer too. The lowest sending speed is 10 WPM.

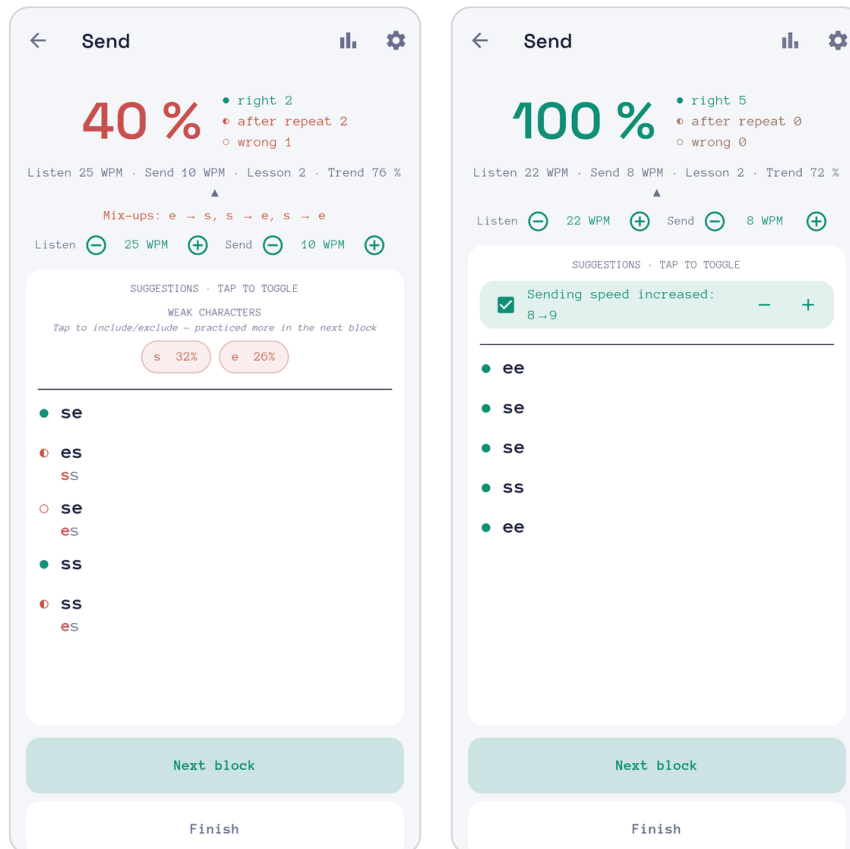
With the **straight key** (keyer mode Straight) there is no set sending speed: the app measures it from your keying (see below). The **Send** row is then disabled and shows the measured speed.

5.4 The result page

After the last word of a block the result page appears:

- **Accuracy** is the share of words correct on the **first attempt**. The colours are the same as in Listen.
- Next to it is the breakdown:
 - ● **right**: correct on the first attempt.
 - ◐ **after repeat**: correct only after a repeat.
 - ○ **wrong**: not made even after all repeats.
- The **status line** shows listening speed, sending speed (if capped), lesson and the trend.
- **Mix-ups** lists which characters you confused in this block, as “target → given”, for example p → w. A — means nothing came at that position.
- The **Listen** and **Send** speed controls. A change here replaces the matching suggestion.
- **Suggestions** and **Weak characters** (see Adaptive mode).
- The list of all words in the block. The target word is shown in bold, with your **first** attempt below it and the first wrong character highlighted in red. A — means nothing came at that position.

Next block applies the ticked suggestions, boosts the ticked weak characters in the next block, and starts it. **Finish** applies the ticked suggestions without a boost and returns to the start view.



Result page: breakdown, mix-ups, weak characters, first attempts

A suggestion (here: raise the sending speed), ticked

5.5 Settings in the ⚙ sheet (Send)

The **Koch sequence**, **Practice set**, **Spacing** and **Word selection** sections are the same as in Listen (see Settings in the ⚙ sheet (Listen)), but they apply to the Send profile. There are two differences:

- **Spacing** applies to the word played to you **and to your answer**, as on the Morserino. Your answer counts as finished once you pause this long after a character:

$$2 \times \text{character spacing} + 1 + \text{word spacing} / 8 \quad (\text{dits at the answer speed})$$

With a straight key it is word spacing + 1 dits, measured at your own speed (up to 30 WPM; above that the shorter straight-key gaps apply). Example: character spacing 28, word spacing 40, answer speed 18 WPM gives 62 dits, about 4 s. That is how long you may pause between the characters of a group, and how long the app waits after the last character before it scores. If that is too lenient or too slow for you, lower the character spacing; the normal 3 / 7 gives 8 dits. When the adaptive feature shortens the spacing, the answer gets stricter too. Longer spacing also gives you more time to begin your answer.

- **Word selection** also uses the call sign settings from the global settings (see Call signs).

In addition there is the **Echo Trainer** section:

Setting	Meaning	Values
Think time	Extra time to begin your answer	1–20 s (8 s)
Repeats	How often a wrongly answered word is played again before the app reveals it. “Forever” repeats until you get it right	0–6 (3), Forever
Prompt	How the prompt is given. Sound means you only hear it. Display means you only read it, with no audio. Both means you hear it and then read it once it has played	Sound / Display / Both
Sending speed (max)	Highest speed for your answer, see Sending speed	same as prompt , 10–50 WPM
Tone shift	Your sidetone while answering is half a tone above or below the prompt	No shift / Up ½ / Down ½
Confirmation tone	Short tone after grading: high for right, low for wrong	Off / On

Prompt = Display is a good way to get from written text to sending, for example to drill new characters.

5.6 Practicing a single character

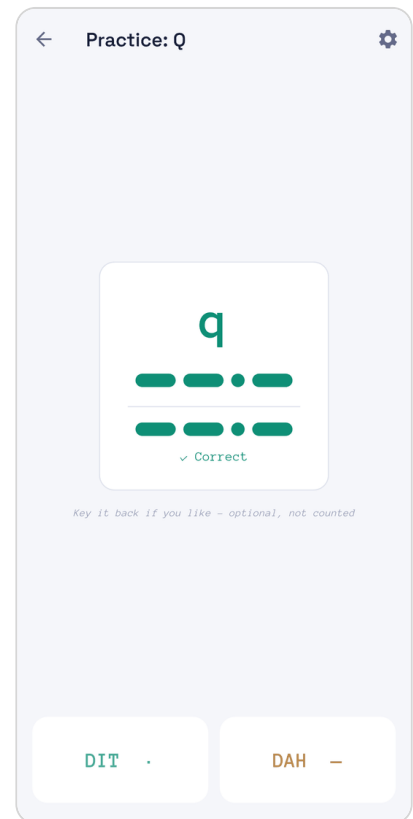
In Listen or Send, long press a Koch character (locked ones too) to open **Practice: X**. In the middle is the same tile as for a tap: the character and its Morse code, whose dits and dahs light up while it plays. The character plays over and over.

After each play you **can** key the character back, with the touch paddle at the bottom or a connected key, but you don't have to. Your dits and dahs appear below the line in the tile as you key them. Once the character is complete they turn green (✓ **Correct**) or red (✗ **You keyed:** with the character you keyed), and shortly after the character plays again. If you key nothing, it repeats after a pause. That is no error, and nothing is counted or added to any statistics. **Back** leaves the page.

Send * sheet: the Echo Trainer section

Set the length of the pause with the **gear** icon at the top right:
Pause before repeating, 1–20 s (default 4 s). It applies only to this practice page, not to Send.

This corresponds to “Learn New Chr” or “Preview Char” on the Morserino.



Practice: a character, keyed back correctly

6 Adaptive mode

Listen and Send always run in **blocks**. After each block the app evaluates how it went and gives you **suggestions**. It may suggest unlocking the next Koch character, making the pauses shorter or longer, or raising the speed. **None of this happens behind your back**. Every suggestion appears on the result page, and you decide whether to accept it.

This chapter explains exactly how the app calculates. You don't need it to practice, but it helps you understand the suggestions and set the thresholds deliberately.

6.1 What the app tracks

6.1.1 Per character

The app keeps statistics for every character, separately for **Listen** and **Send**:

- **attempts**: how often the character was graded,
- **errors**: how many of those were wrong,
- **moving error rate**: the basis for all decisions.

The **moving error rate** is an exponential moving average (EMA). On every attempt:

new rate = $0.2 \times (1 \text{ on error, else } 0) + 0.8 \times \text{old rate}$

Each new attempt counts 20 % and the past counts 80 %. Recent results weigh more than old ones, but a single slip doesn't upset everything. For example, a character at 0 % error rate that gets one error is at 20 %. It then takes 3 correct attempts to get back below 12 %, and 8 correct attempts to get below 5 %.

A character's **accuracy** is 100 % minus its moving error rate.

What counts as an attempt?

- **Listen**: every character played in a block. It is correct unless you marked it as an error. When copying on the on-screen keyboard, the app marks the errors of the **first** attempt of each word itself (see Copying on the on-screen keyboard).
- **Send**: only the **first** attempt of each word. The characters before the first error count as correct, and the first wrong character counts as an error. The characters after it aren't counted, because it's unclear whether you heard them correctly. Repeats of the same word don't count.

6.1.2 Per block

Every block has a **block rate**:

- **Listen**: characters copied correctly / all characters.
- **Send**: words correct on the first attempt / all words.

From the block rates the app forms another moving average, the **block EMA**:

block EMA = $\alpha \times \text{block rate} + (1 - \alpha) \times \text{previous block EMA}$

α is the **EMA smoothing** setting (default 30 %). The block EMA starts at 100 %. It is kept across sessions, separately for Listen and Send.

6.2 The two thresholds

With **Success threshold low/high** (default 70 % / 90 %) the app divides the block EMA into three ranges:

Block EMA	Meaning	Suggestion
90 % and above (high)	Going well	After 2 blocks in a row in this range: shorten the pauses, or raise the speed if the pauses are already normal
70 % to below 90 %	About right	Change nothing
below 70 % (low)	Too hard	Immediately: lengthen the pauses

The high threshold is also the accuracy every character has to reach before the next Koch character unlocks.

6.3 When the next Koch character comes

The next Koch character is suggested when **every** character of the current lesson meets two conditions. That means all of them, not just the most recently learned one:

1. at least **20 attempts** (the **Occurrences for unlock** setting), and
2. an **accuracy of at least 90 %** (the high success threshold).

Two points matter here:

- The condition applies to each character individually. A single weak or rarely practiced character holds up the unlock. The progress card and the character statistics show you which one it is.
- Because accuracy is a moving value, it isn't enough to have got a character right 20 times at some point. Your **recent** attempts have to be good.

The unlock appears as a highlighted suggestion with a star: **New character unlocked: “X”**. The  icon next to it plays the new character three times, without leaving the result page. It shows the same tile as tapping a Koch character: the character with its code below, each element lighting up as it sounds. A tap closes it early. If you untick it, you stay in the current lesson.

6.4 Pauses and speed

The app always changes the **pauses** first, and only then the speed:

- **Shorten:** character spacing and word spacing each get 1 dit shorter, down to the normal 3 / 7 dits.
- **Raise the speed:** only once the pauses are already at 3 / 7 dits does the app suggest **+1 WPM**.
- **Lengthen:** character spacing and word spacing each get 1 dit longer, but never longer than at the start of the session. In Listen that is when you opened the training. In Send it is the value from the * sheet.

When copying on the **on-screen keyboard**, the app only changes the **character spacing**; the word spacing stays, because the app waits for you after every word there anyway. The pauses then count as normal as soon as the character spacing is at 3 dits.

While you are still working through the Koch sequence, the app neither shortens the pauses nor raises the speed. You should be able to concentrate on new characters without everything getting faster at the same time. Lengthening is always possible. Shortening and raising the speed only start once one of these is true:

- all characters of the Koch sequence are unlocked, or
- you practice with **All characters** or a **Practice set**.

The block in which a new character is added also never shortens the pauses or raises the speed.

You can step in yourself at any time, independently of these rules. In Listen use **Adjust spacing**, and in Send use the speed controls.

6.4.1 Sending speed in Send

Send has one extra suggestion: **Sending speed increased (+1 WPM)**. It only appears if you have set a sending speed **below** the listening speed and the block reached at least the high threshold. It starts **unticked**, because the sending speed is a deliberate choice. With the straight key this suggestion is dropped, and so are the tighter/wider spacing suggestions; listening speed and new characters are still suggested.

6.5 Accepting, rejecting, adjusting suggestions

Each suggestion is a row on the result page:

- The **checkbox** on the left accepts or rejects it. Most suggestions start ticked.

- – and + adjust the size. The speed goes up to at most 5 WPM above the current one. The pauses can be anywhere between 3 / 7 dits and the value at the start of the session.
- The status line immediately shows the values for the next block.

Nothing is applied until you leave the result page with **Next block** or **Finish**.

6.6 Weak characters

A character counts as **weak** if both of these are true:

- it has at least **8 attempts**, and
- its moving error rate is at least **12 %**.

At most the **5** weakest are shown, worst first, with their error rate. Because the rate is tracked over time, a character stays weak until you get it right reliably again. This can take several blocks and sessions.

Tap a weak character to exclude it from the boost (it appears struck through), or to include it again.

- **Listen:** the weak characters already appear on the start view and on every result page. In the next block they are boosted at the *Moderate* level, with up to 3 draws per character (see Practice set). This only applies to random groups (Koch lesson or All characters · Random), not to words. If you have your own practice set characters and **Boost practice set** switched on, they are added, and the higher level applies (so Strong for all of them if you chose Strong).
- **Send:** the weak characters appear on the result page when you practice **Koch lesson · Random**. With **Next block**, the ticked characters come up twice as often in the next block. The boost lasts exactly one block.

6.6.1 Weighting in Send

In **Send · Koch lesson · Random** the app also draws the characters by weight. On the Morserino this is called “Adaptive Random”. Every character has a weight between 1 and 20, and the higher it is, the more often the character comes up. After the first attempt of each word:

- **Word fully correct:** each character of the word gets –1.
- **Error:** the first wrong character gets +4, and its neighbours in the word get +2 each (if they are different characters).

Characters you get wrong come up more often quickly. They only become rarer again once you key them correctly several times. That isn’t always pleasant, but it’s very effective.

6.7 Progress card “On the way to X”

On the **Listen** result page with the Koch lesson, a card shows what is still missing before the next character:

- The **bar** shows how many of the required attempts you have made, added up over all characters that still lack attempts. If no attempts are missing any more and only accuracy is, it shows the weakest character’s accuracy relative to the threshold.
- **Still to practice (repetitions)** lists characters that still lack attempts, each as a small chip with the number missing, for example q 7. Those missing the most come first.
- **Accuracy below 90 %** lists characters with enough attempts but too low an accuracy, for example y 84 %. The weakest come first.

Each list shows at most 10 characters; the rest appear as “+N more”.

6.8 Trend

From the **sixth** block on, the status line shows a trend, for example “Trend 87 % ▲”:

- The number is the average block rate of the **last 5 blocks**.
- The arrow compares it with the average of the 5 blocks before. ▲ means at least 3 percentage points better, ▼ means at least 3 points worse, and ► means the same.

The trend is kept across sessions, separately for Listen and Send. The app stores the last 20 blocks for it.

6.9 Adaptive mode settings

These are in the * sheet of **Listen** under **Adaptive mode**. They apply to Listen and Send together.

Setting	Meaning	Values
Success threshold low/high	Below <i>low</i> the pauses are lengthened. At <i>high</i> (2 blocks in a row) they are shortened or the speed is raised. <i>High</i> is also the accuracy each character needs for the Koch unlock	30–99 %, at least 5 points apart (70 % / 90 %)
EMA smoothing	How strongly the last block counts in the block EMA. Higher reacts faster, but is also jumpier	5–100 % (30 %)
Occurrences for unlock	Minimum attempts per character before the next Koch character can come	5–50 (20)

Some notes:

- The high threshold can’t be set to 100 %. The moving error rate of a character that ever had an error never quite returns to 0. At 100 % that character would block the unlock forever.
- **EMA smoothing 100 %** means only the last block counts.

- If unlocking goes too fast for you, raise **Occurrences for unlock** or the high threshold. If it's too slow, lower them.

7 Character statistics

The  icon in Listen and Send opens that training's statistics. Listen and Send have separate statistics.

- **Listen** shows every active Koch character with its attempts (for example “14 attempts”) and its **current** accuracy. A tick marks characters that meet the unlock condition, and an hourglass marks those that aren't there yet. Those also say what is missing (“6 more needed”, “below 90 %”). The top shows “x of y characters ready”. The characters that aren't ready come first, which helps when the unlock seems stuck.
- **Send** shows the same (attempts, current accuracy, tick or hourglass, “x of y ready”), but the characters are sorted by errors, the least reliable at the top. The unlock applies here too, for Koch with random characters. Mix-ups are not listed in the overview; you find them per character in the detail view.

The percentage is a moving average, not an overall rate. Newer attempts count more than older ones: each attempt goes into the value with 20 %. A miss therefore lowers it at once (100 % becomes 80 % at most), and correct answers rebuild it slowly. A character can drop below the threshold after a single miss even if it was flawless before. The tick needs both: enough attempts (default 20) **and** a current accuracy above the threshold (default 90 %). The bar shows only the attempts, not the accuracy.

Tap a character to open its detail view:

- **Overall** – accuracy over all attempts, with errors and attempts.
- **Current (moving)** – the value from the list, with an arrow: ▲ better, ▼ worse or ► same as the overall rate (from 3 percentage points apart, and only from 5 attempts). It shows whether a dip was a slip.
- **Last 30 attempts** – a strip of green (right) and red (wrong) bars, newest on the right. It fills up as you practise; older data has no history.
- **Hit rate per week** – a curve of the last 12 weeks for this character (tab *Hits*; dashed: the unlock threshold) or the number of attempts per week (tab *Attempts*). Weeks without practice are left out.
- **Last practised** – today, yesterday or *n* days ago (also only from the next practice on).
- **Practice weight** – from 1 to 20. The higher, the more often the character comes up in adaptive exercises.
- **Unlock** – what is still missing: how many attempts and how many right answers in a row until the rate is back above the threshold.
- **Mix-ups** (Send only) – what you sent instead. In Listen you only mark what was wrong, so what you heard is not recorded there.
- **Listen** – plays the character three times, at this training's speed.



Detail view in Listen: weekly curve, unlock

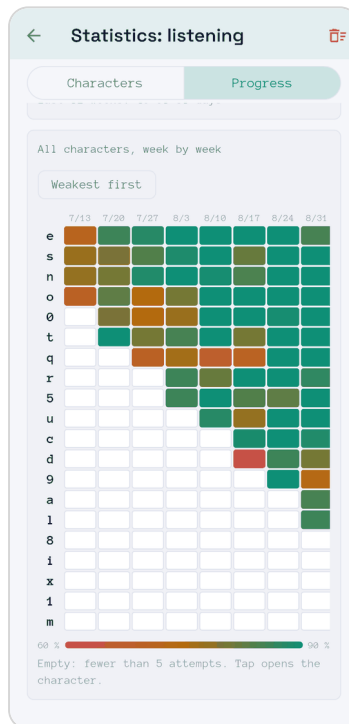
Detail view in Send: last 30 attempts, curve, mix-ups

Progress tab. At the top, switch between *Characters* (the list above) and *Progress*. Progress shows how practice develops over time, separately for each training. It fills up from the day this feature was added; earlier practice is not included. Choose **4 weeks**, **12 weeks** or **All**:

- **Hit rate, WPM and days practised** at the top. Hit rate and WPM are those of the latest day (4 weeks), week (12 weeks) or month (All, from about half a year on) with practice, weighted by attempts, with an arrow against the period before. “Days practised” reads like 42/88.
- **Hit rate per day / week / month and speed (WPM)** as curves. Periods without practice are gaps, not zero.
- **Days practised:** one box per day (4 weeks), otherwise a bar per week or month.
- **All characters, week by week:** a heatmap, characters down, weeks across, from red (below 60 %) to green (from 90 %). A cell with fewer than 5 attempts stays empty. The characters and the week headings stay in place while the cells scroll. **Weakest first** sorts by the latest hit rate. Tap a character to open its detail view.



Progress tab (Listen): hit rate, speed and days practised over 12 weeks

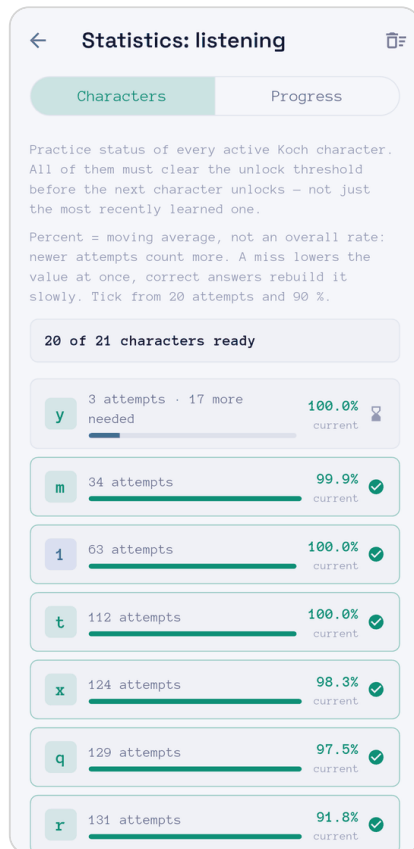


Below it the heatmap: all characters, week by week

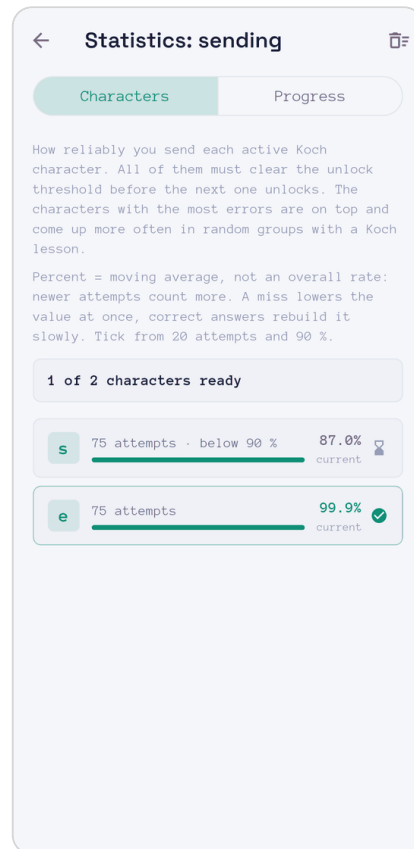


Progress tab (Send)

The **reset** icon deletes all of this training's statistics after a confirmation: error rates, weights, mix-ups and the progress. The other training's statistics are kept. This cannot be undone.



Listen statistics: attempts, accuracy, ready



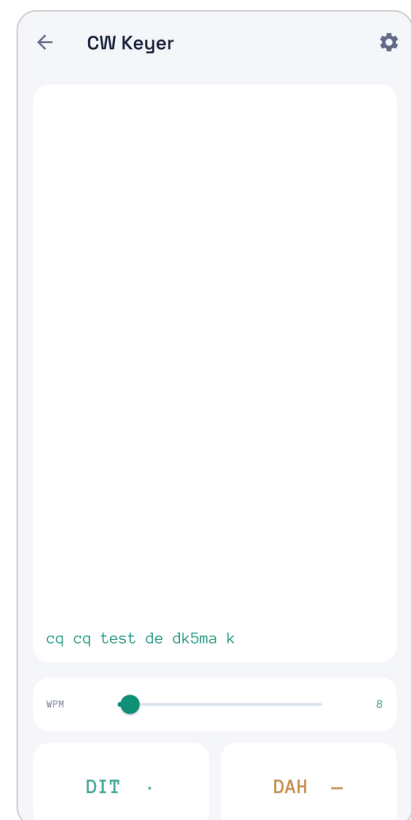
Send statistics with common mix-ups

8 CW Keyer

For free keying. What you key is heard and shown as decoded text.

- Use the **paddles** at the bottom (DIT on the left, DAH on the right), or a real paddle through an adapter (see Paddle and Morse key). In **Straight** mode a single **KEY** area appears instead.
- **WPM** sets the keyer speed, 5 to 60 WPM.
- The text runs from the bottom upwards. You can scroll back to older lines, and you change the text size with two fingers.
- The * sheet at the top right holds the **word spacing** (word spacing, default 7 dits). It sets the pause after which a space is inserted. Character spacing has no effect when keying, as on the Morserino.

You set the keyer mode and its details in the global settings under **Keyer** (see Keyer). They apply everywhere you key: CW Keyer, Send, WiFi Trx, QSO Bot and the games.



CW Keyer with decoded text

9 CW Decoder

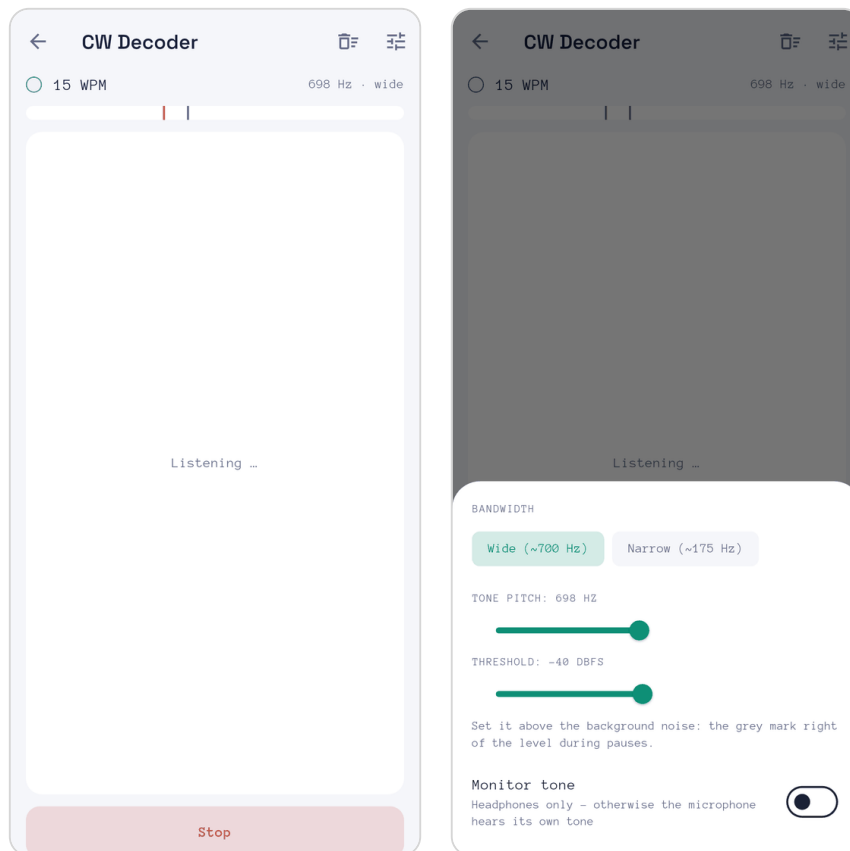
The CW Decoder listens through the **microphone** and writes down what it recognises as Morse code. The signal might come from a radio, a practice program or another Morserino.

On first start Android asks for microphone permission. Without it the decoder can't work. If you declined, allow the microphone in the Android settings under **Apps → Next CW Trainer → Permissions**.

9.1 Operation

- **Start / Stop** switches copying on and off.
- The **status line** shows the set pitch, the bandwidth and the detected speed in WPM.
- The **level meter** shows the loudness of the tone at the set pitch. A grey mark shows the noise level during pauses.
- The icons at the top right clear the text and, with the sliders icon, open the decoder settings.

The decoder adapts to the speed automatically, so there's nothing to set. It comes from the firmware. A Goertzel filter detects the tone, and timing logic tells dits, dahs and pauses apart. It keeps learning the current dit and dah length as it goes.



CW Decoder listening, with level meter

Decoder settings

9.2 Decoder settings

Setting	Meaning	Values
Bandwidth	Wide is more tolerant of a pitch that doesn't quite match. Narrow filters interference better, but the pitch must be exact	Wide (~700 Hz) / Narrow (~175 Hz)
Tone pitch	The frequency of the CW tone the decoder listens for	300–1200 Hz (698 Hz)
Threshold	Minimum loudness for a tone to count. Set it just above the grey noise mark	–65 to –10 dBFS (–40 dBFS)
Monitor tone	Plays back the detected tone cleanly at the set pitch	Off / On

Tips

- If you don't know the signal's pitch, start with **Wide** and turn the pitch until the level jumps clearly on the characters. Then switch to **Narrow**.
- Random **e** and **t** appearing in the pauses mean the **threshold** is too low.
- Only use the **monitor tone** with headphones. Otherwise the microphone hears the tone itself and confuses the decoder.

10 WiFi Trx

With WiFi Trx you send Morse over the internet or your local network with other Morserinos, apps and servers. The app uses the same protocol as the Morserino (“Morse over Packet”, MOPP, UDP port 7373). One example is the chat server **cq.morserino.info**.

The app uses the phone's existing WiFi or mobile data connection, so there are no WiFi credentials to enter.

10.1 Services

At the top you choose the **service** to connect to. The default is **cq.morserino.info**.

- **+** adds a new service, and the **pencil** edits or deletes the selected one. Each service has a **Name** and a **Server** (host name or IP address).
- An **empty server** means broadcast to the local network. This lets you exchange Morse with Morserinos on the same WiFi.
- You can only do this while not connected.

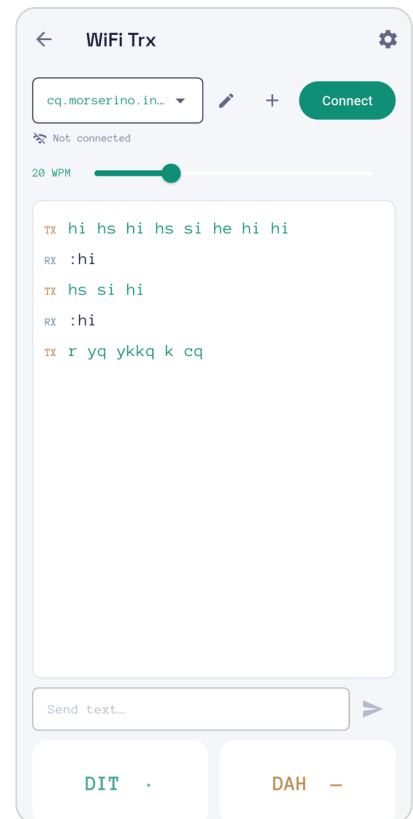
Connect connects and **Disconnect** disconnects. The line below shows the status. If the server ends the connection (**:bye**), the app disconnects too.

10.2 Sending and receiving

- **Send** with the paddles or the adapter. Each word is sent as a packet after the word pause. You can also type text into **Send text...** and send it. It goes out at the set speed.
- **WPM** sets your sending speed.
- **Receiving:** received words are played one after another at the **sender's** speed and shown in the log.
- The **log** tells received (RX) from sent (TX) text and is kept per service. **Long-press the log** to clear it after a confirmation.
- In the ***** sheet you set the **word spacing**, as in the CW Keyer. It decides after which pause your word is sent.

WiFi Trx only works while the app is in the foreground.

If your partner is behind a different router, UDP packets on port 7373 must be able to get through. With a public server such as cq.morserino.info this is normally no problem.



WiFi Trx with service, log and text field

11 QSO Bot

The QSO Bot is a simulated partner for complete, realistic radio contacts, without going on the air. The bot sends in CW, understands what you key and reacts to it. It only sounds through the app's speaker and never sends anything to the network.

11.1 Starting

1. Choose the **QSO type: SOTA/POTA, Standard or Contest**.
2. Press **Start**. The bot listens for 5 seconds:
 - If you call **CQ** (for example `cq cq de oe1abc k`), it answers you.
 - If you stay silent, it calls CQ itself and you answer.

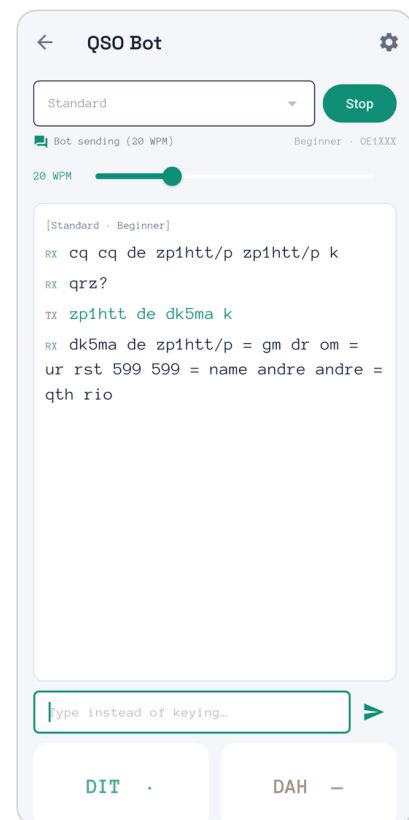
Each contact uses a new, realistic call sign for the bot.

11.2 QSO types

- **SOTA/POTA** is a summit or park activation with report and reference. If the bot calls CQ, it is the activator and you are the chaser. If you call CQ, it's the other way round. A session is a single contact.
- **Standard** is the classic QSO in three rounds. First come report, name and QTH. Then station details follow (rig, ant, wx, age), and finally the sign-off. The bot finds your details by the keywords `name`, `qth`, `rig`, `ant`, `wx` and `age`, and it remembers your name.
- **Contest** is many very short contacts in a row. The exchange depends on the **Contest type: CQ WW** (report + CQ zone) or **WPX** (report + serial number). The session ends by itself after a longer silence.

11.3 Talking to the bot

- **End each over** with `k`, `bk`, `<ar>`, `<sk>` or `73`, or simply pause. The bot won't talk over you.
- **Repeat:** `agn`, `rpt` or `?` repeats the bot's last over. `rpt rst`, `rpt call`, `rpt qth` or `rpt name` repeats just that item.
- **Bot speed:** `qrs` makes it slower and `qrq` faster. Your own speed doesn't change.
- **Correcting:** `<err>` (eight dits) or `eeee` discards your last word.
- You can send reports in cut numbers: `5nn` for 599, `t` for 0, `a` for 1, `n` for 9.
- Filler words like `de`, `r` and `ur` don't matter, because the bot picks out the information.



A QSO with the bot, answered with the text field

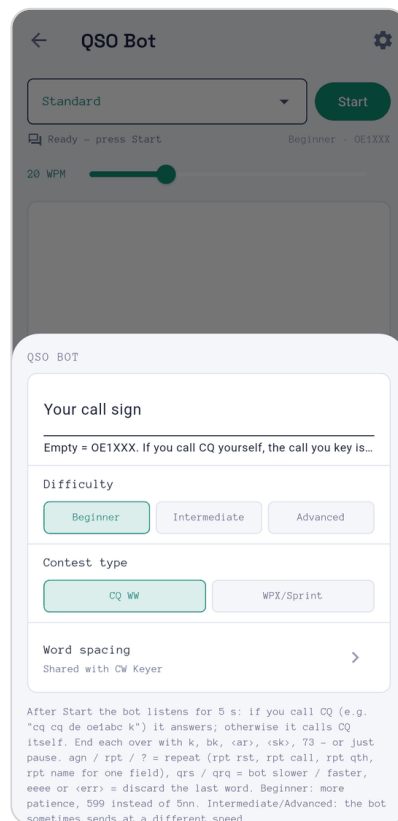
- Instead of keying, you can type text into the input field. It is treated as if keyed, but not played aloud.

11.4 QSO Bot settings

These are behind the ⚙ icon in the QSO Bot:

Setting	Meaning	Values
Your call sign	Your call sign for the bot. If you call CQ yourself, the call you key is used	empty = OE1XXX
Difficulty	Beginner: more patience, 599 instead of 5nn, and the bot sends at your speed. Intermediate and Advanced: the bot sometimes calls at a slightly different speed, as practice for qrs/qrq. Advanced also keeps a tighter rhythm	Beginner / Intermediate / Advanced
Contest type	Exchange in a contest	CQ WW / WPX
Word spacing	After which pause your word counts as finished. It is shared with the CW Keyer	6–105 dits (7)

WPM sets your speed. **Long-press the log** to clear it after a confirmation.



QSO Bot settings

12 Own texts

Own texts lets you listen to any text as Morse code – a weather report, a newspaper article, a book chapter or a list of call signs. You find it under **Free modes**.

12.1 Adding texts

1. In any app, copy a text to the clipboard.
2. In **Own texts**, tap **Paste from clipboard**.
3. Give the text a title (the first words are suggested) and tap **Add**.

You can add as many texts as you like. A text can be at most 20,000 characters long. The **:** menu next to a text renames or deletes it (after asking). You can't edit a text in the app: to change one, paste the corrected version. The texts stay on your device.

12.2 What is sent

The text is always shown as you pasted it. Only the audio replaces characters Morse doesn't have:

- Umlauts and ß: Ä → AE, Ö → OE, Ü → UE, ß → SS. Other accents are dropped (é → E).
- **!** becomes **.**, **;** becomes **,**, **&** becomes **AND**, dashes become **–**. A dash standing alone between spaces is skipped.
- Quotation marks, brackets and other special characters are not sent.
- Write **prosigns** in angle or square brackets, e.g. **<KA>** or **[SK]**; they are sent as one character. A pair of letters without brackets is always two letters.

A paragraph break in the text is sent as a double word gap.

12.3 Playing

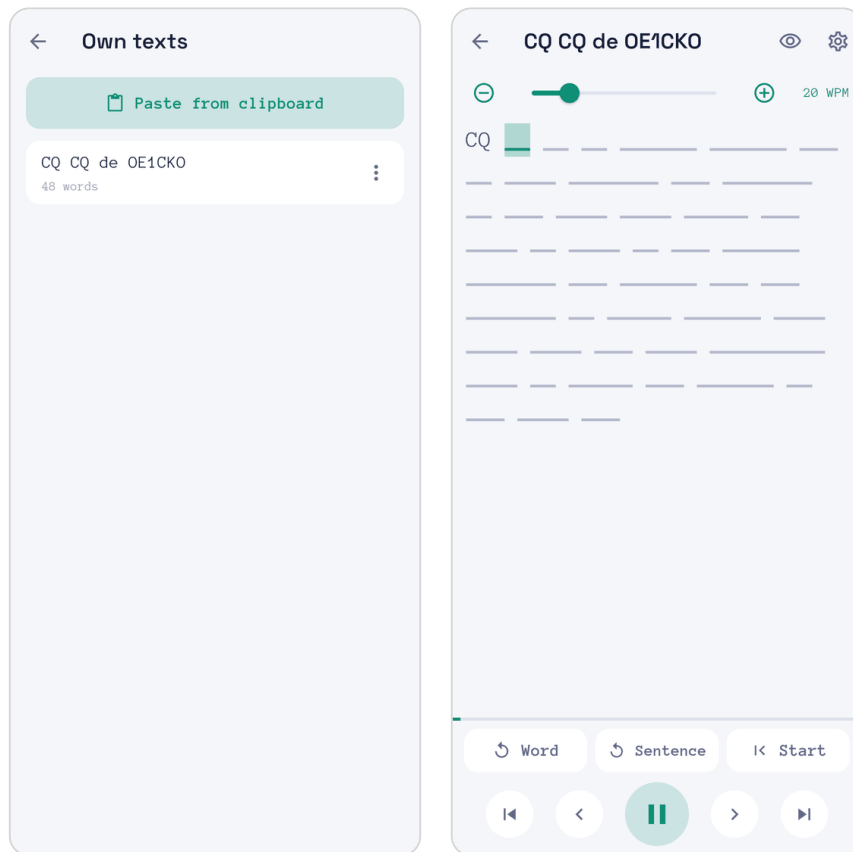
Tap a text, then **▶**. At the top you set the **speed** (WPM) with the slider or **⊖/⊕** – also **while** the text plays; the change applies at once.

Button	Effect
▶ /	Play; pause (then it goes on from the start of the current word)
Word	Repeat the current word, then pause
Sentence	Repeat the current sentence, then pause
Start	The whole text from the beginning
⏮ / ⏭	To the start of the previous / next sentence

Button	Effect
< / >	One word back / forward

Tap any word in the text to listen from there. Pinch with two fingers to make the text larger or smaller. At the end the text starts again from the beginning.

Progress is saved automatically. When you open a text you have started, the app asks: **Continue** or **Start over**. The list shows how much of each text you have heard.



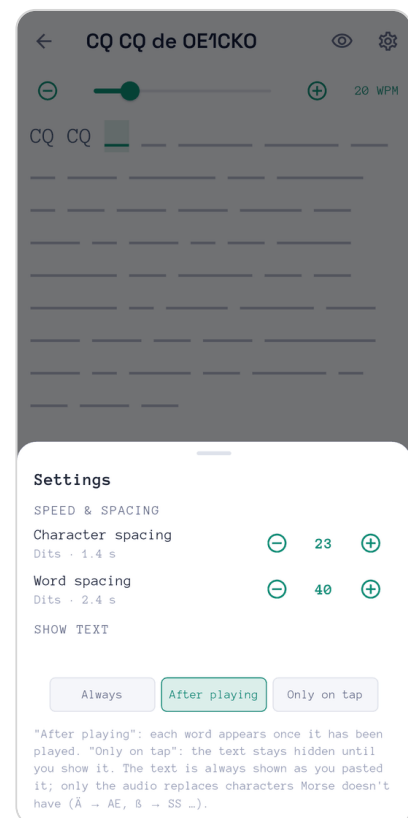
Own texts: the library

A text playing (display: After playing)

12.4 Settings (⚙️)

- **Character spacing** and **word spacing** in dits, as in the trainings. They also apply while playing.
- **Show text: Always; After playing** (each word appears once it has been sent – the default); **Only on tap** (the text stays hidden until you show it with the eye icon at the top).

Speed, spacing and display apply to all texts. Pitch and tone softness come from the general settings. The display stays on while you are in a text.



* sheet of Own texts

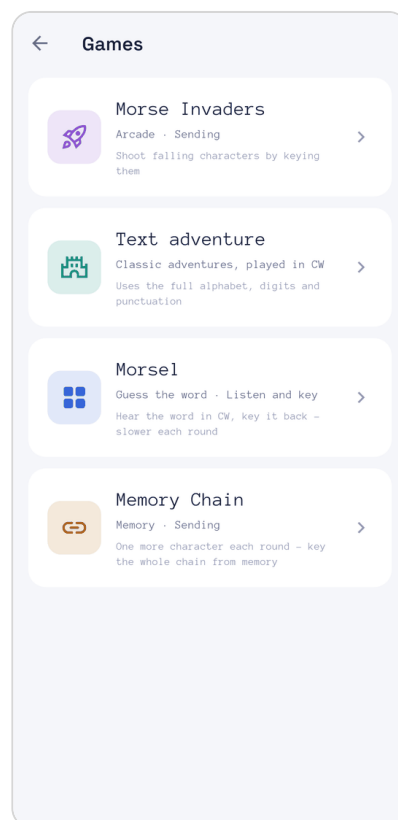
13 Games

Under **Play → Games** there are four games. You play Morsel, Morse Invaders and Memory Chain with the paddles or the adapter. They use the keyer settings. They take the Koch lesson from **Send**, and Morsel and Memory Chain also let you change it for the game only. Each of these three shows short rules before you start. The text adventure has its own speed and input settings; it also takes the keyer mode from the keyer settings.

13.1 Morsel

A word puzzle like Wordle, but in CW.

- A game has **ten words**. One letter is revealed, and the word is played once in CW.
- Key the **whole word** back. A word pause submits it.
`<err>` (eight dits) deletes the last character.
- Colours: **green** means right, **red** means wrong, and **grey** means the revealed letter was keyed wrong.
- After each miss the word is repeated **5 WPM slower**, down to 18 WPM. After 12 seconds without input it is played again. **Skip** gives up on a word.
- **Score**: your time + 5 seconds per guess. A skipped word costs 60 seconds. Lower is better.



The games

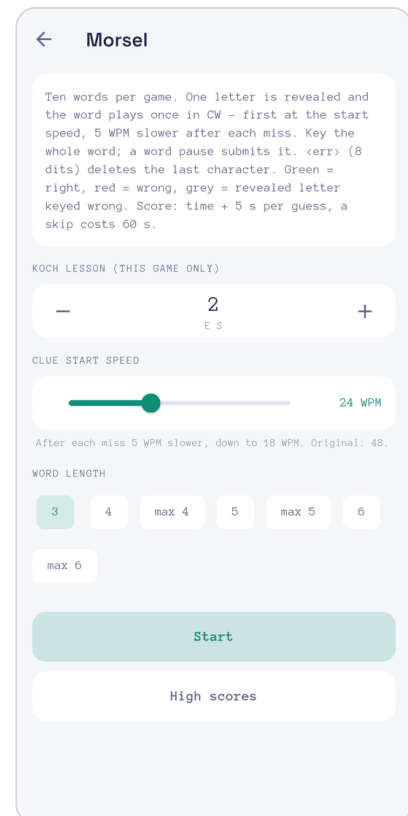
Setting	Meaning	Values
Koch lesson	For this game only. It starts at your Send lesson	2 to the end of the sequence
Word length	Exactly 3/4/5/6 letters, or at most 4/5/6	3, 4, max 4, 5, max 5, 6, max 6
Clue start speed	Speed of the first play	10–48 WPM (48 , as on the Morserino)

Longer words need a higher Koch lesson. If too few words are available, the app tells you which lesson you need.

13.2 Morse Invaders

An arcade game. Characters from your Koch lesson fall down four lanes. **Key a character** to shoot the lowest invader showing it.

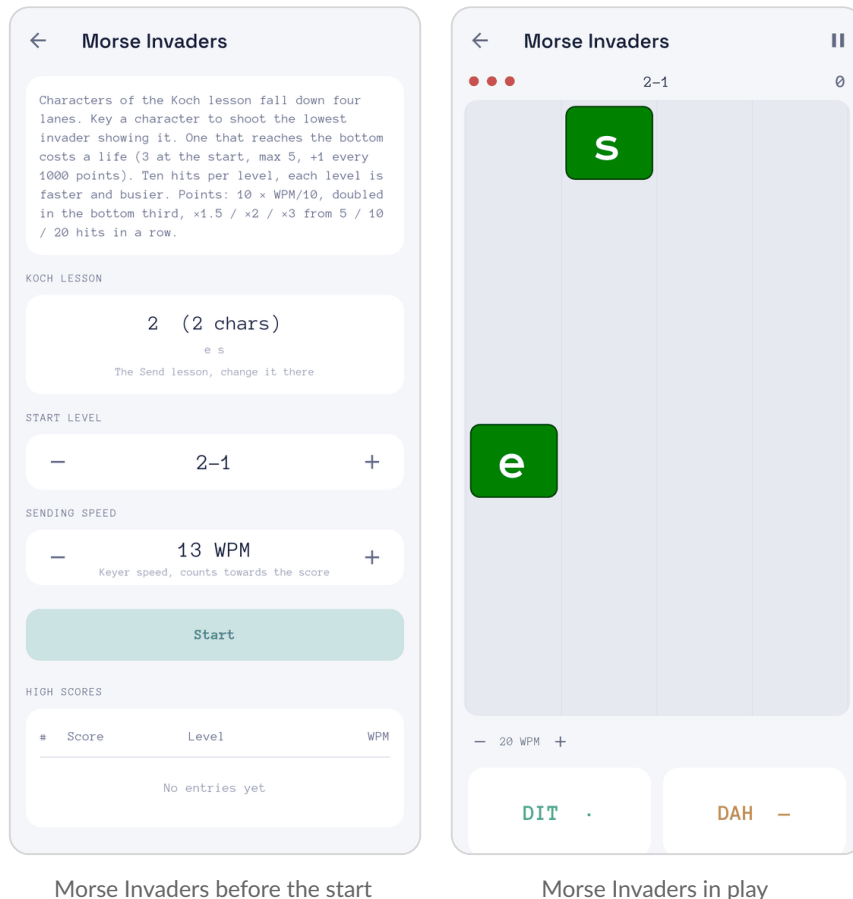
- A character that reaches the bottom costs a **life**. You start with 3 lives, can have at most 5, and gain one every 1000 points.
- After **ten hits** comes the next level. Each level is faster and busier.
- **Points:** $10 \times \text{WPM} / 10$ per hit, doubled in the bottom third. Streaks add a multiplier: $\times 1.5$ from 5, $\times 2$ from 10 and $\times 3$ from 20 hits in a row.
- **Pause** (the icon at the top right) stops the game. From there you can resume or go back to the menu. You can also change the speed with – and + during the game.



Morsel before the start

Setting	Meaning
Koch lesson	The lesson from Send (change it there)
Start level	The level you begin with
Sending speed	Keyer speed, counts towards the score

Results go into a high score list.



Morse Invaders before the start

Morse Invaders in play

13.3 Memory Chain

A memory game. Each round adds **one character**, and you key the **whole chain** from the start, from memory. There is no time limit.

- **Characters mode:** random characters from your Koch lesson. One error per round is allowed, and the second ends the game.
- **Call signs mode:** a call sign is built up letter by letter, then the next one. Any error ends the game.
- **Prompt:** the new character is **shown** or **played** in CW (Sound).
- **Colours:** **green** means right, a **yellow frame** marks the next character, and **red** marks an error (with the right character).
- You set the **Koch lesson** for this game only before you start. It begins at your Send lesson. You can also change the speed with – and + during the game.

The high score list is kept per mode.

13.4 Text adventure

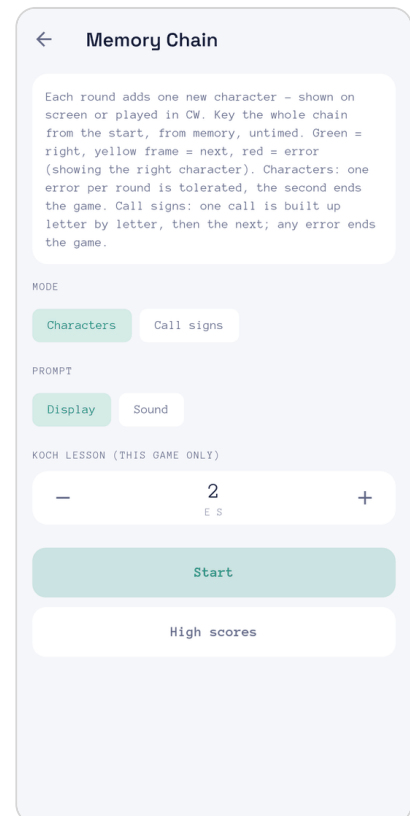
The three classic Infocom adventures **Zork I, II and III** (1980–82), played in CW: the game answers in Morse, and you enter your commands. Microsoft released the source code under the MIT License in 2025; the app plays the original story files with its own interpreter. The game text is in **English** only, and it uses the **full alphabet, digits and punctuation**, whatever your Koch lesson. Zork is a trademark of its owners; the app is not affiliated with them.

Selection. Each part shows its score, moves and last room. **Start** or **Continue** picks up where you left off. **Saved games** opens your saves. **New** starts over (after asking; your own saved games are kept).

The game screen. At the top the status line with room, score and moves, below it the transcript. The newest answer is played in CW, and the word that is sounding is highlighted. Below that:



Choosing one of the three parts

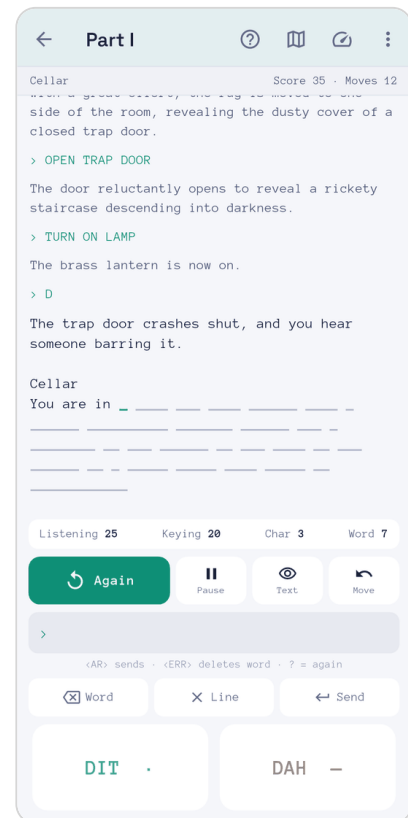


Memory Chain before the start

Control	What it does
Speed bar	Listening and keying speed, character and word spacing; tap it for Speed & spacing
↺ Again	The sentence that is playing (in the gap right after a word: that word's sentence; after the end: the last one), then paused
Long press ↺ Again	The whole answer from the start (like sending ?)
Tap a word	Only that word, then paused
Pause / Resume	Pauses playing; Resume goes on from the word where it paused – after ↺ Again from the next word
Text (eye)	Shows the whole answer at once; tap again to hide it, then the words reappear as they are played (e.g. with Again)
↶ Move	Takes back the last command (up to 20)

Keying commands. With the touch paddles at the bottom (keyer mode Straight: the key) or a connected paddle (see Paddle and Morse key). The decoder writes into the input line; the character being keyed is shown in orange as · and — after it. As soon as you start keying, the CW output stops.

- **End of a word:** a pause separates words. As in the sending learn mode, a word counts as finished after a pause of $2 \times \text{character spacing} + 1 + \text{word spacing} / 8$ dits at keying speed (straight key: word spacing + 1 dits after key-up).
- **Sending:** <AR> (·—·—·) sends the command at once. With the setting “or K”, a K as its own word also sends, i.e. after a word gap – none of the three games has a word K. The **Send** button always works.
- **Correcting:** <ERR> (8 dits; 7 or more count) deletes the last word, as does the ☒ **Word** button. ✕ **Line** clears the whole input.
- A ? on its own (then send) repeats the last answer and takes no move.
- Other prosigns are ignored; a character that isn't recognised shows as *.



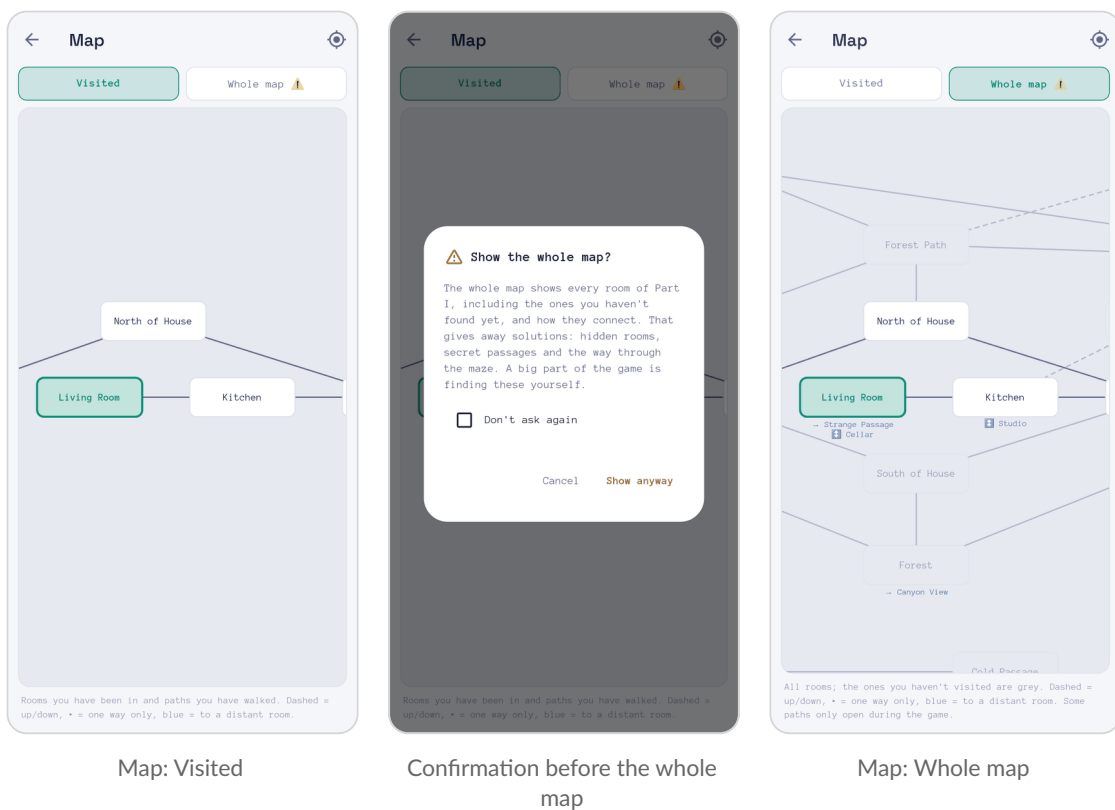
Game screen with touch paddles

With the setting **Input → Keyboard**, the on-screen keyboard replaces the paddles (for listening practice only): _ separates words, ☒ deletes a character, ↵ sends. A connected paddle works then too.

Map (📍). The map icon in the title bar opens a map of the current part. It starts at the current room (orange outline); pinch to zoom, drag to move, ⬅ goes back to the current room. Rooms carry their

names from the game (English). Their positions are set by hand, because Zork isn't drawn to scale; the connections come from the story file.

- **Visited** (default): only rooms you have been in and paths you have walked – the map players used to draw on paper. No cheating. The paths are saved with the game and taken back by ↶ **Move**.
- **Whole map** ⚠: every room of the part, including the ones you haven't found yet (grey). That gives away solutions (hidden rooms, secret passages, the way through the maze), so the app asks every time. **Don't ask again** in the dialog turns the question off; turn it back on under : → Settings → **Map**.
- Dashed = up/down, ▶ = one way only. Distant connections (e.g. trap door, chimney) show as a blue note under the room ("→ Cellar") instead of a long line. Some paths only open during the game.
- In Part III the museum rooms appear three times side by side, labelled with the year (948 = the present, 776, 777): they are the same rooms at different times.



Command overview (?). The ? in the title bar opens a list of the most important commands and, under **Playback**, the buttons above; at the bottom, **What it's about** briefly gives the story and goal of the current part, how points and moves are counted and what happens when you die. The short forms work in all three parts:

Command	Meaning
N S E W, NE NW SE SW, U D, IN OUT	Moving
L	LOOK: describe the room again (takes a move)

Command	Meaning
I	INVENTORY: what you carry
Z	WAIT: wait one move
G	AGAIN: repeat the last command
OOPS word	Replaces a word the game did not know
TAKE, DROP, EXAMINE, READ, OPEN ...	Handling things (Zork has no X for EXAMINE)
SCORE, SAVE, RESTORE, RESTART, QUIT	Game commands
<AR>, K, <ERR>, ?	Only in this app: send, send (as its own word, depending on the setting), delete the last word, again

The game reads only the first **6 letters** of a word (EXAMIN is enough). Separate several commands in one line with a full stop: TAKE LAMP. N.

Settings (: → Settings; shared by all three parts):

```

BRIEF,          how much of a room is described
SUPERBRIEF,
VERBOSE

SAVE, RESTORE   save and load through the app's saved
                games

RESTART, QUIT    start over, quit (both ask first)

PLAYBACK
  ◡ tap          the sentence playing (or just
                played), then paused
  ◡ hold         the whole answer from the start
Pause / Resume  pause; Resume goes on from the word
                (also after ◡)
Tap a word      only that word, then paused
Text           show the whole answer; again: hide it

ONLY IN THIS APP
<AR>           ---: send the command
K              as its own word: send (setting "<AR>
                or K")
<ERR>         8 dits: delete the last word
?             ? alone: hear the last answer again
                (no move)

The game reads only the first 6 letters of a word:
EXAMIN is enough.

WHAT IT'S ABOUT
You are standing outside a white house; beneath it lies
the Great Underground Empire. Goal: find every treasure
and put it in the trophy case in the house's living
room. Each treasure scores when you first pick it up
and again in the case; some special places score too -
350 in all.

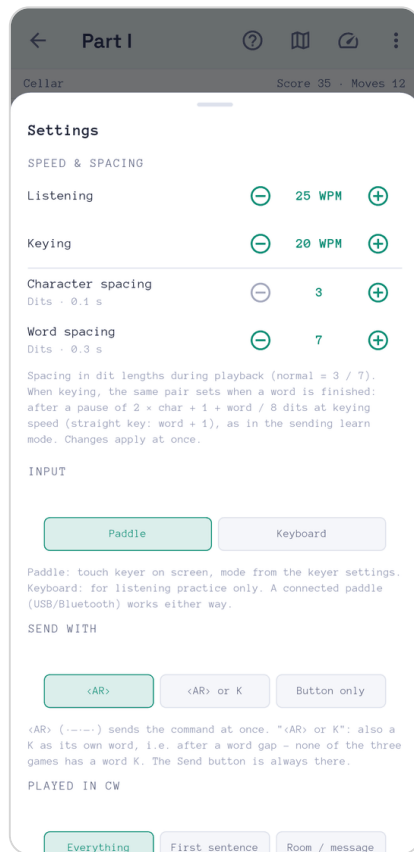
Moves: every command the game understands is a move (in
Parts I and II, SCORE, SAVE, BRIEF and the like don't
count). Time passes with every move: the lamp burns
down, other characters wander about. SCORE shows points
and moves.
If you die, the game usually goes on - in Parts I and
II with 10 points off and your things scattered - but
not forever. : → Undo move takes back the last
command.

```

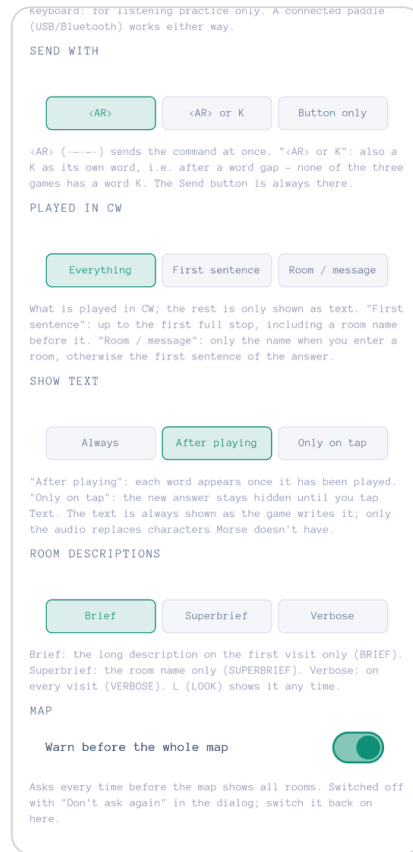
Command overview, "What it's about"
at the bottom

Setting	Meaning	Values
Listening	Speed of the CW output	10–60 WPM
Keying	Keyer speed for your input	as listening , 10–60 WPM
Character spacing	Pause between characters when playing; part of the word end when keying	3–45 dits
Word spacing	Pause between words; never below the character spacing; part of the word end when keying	6–105 dits
Input	Touch paddles or on-screen keyboard	Paddle , Keyboard
Send with	What sends a keyed command (the Send button always works)	<AR> , <AR> or K, Button only
Played in CW	What is played; the rest is only shown as text (italics)	Everything , First sentence, Room / message
Show text	When the new answer becomes readable	Always, After playing , Only on tap
Room descriptions	How much of a room is described	Brief , Superbrief, Verbose
Map: warn before the whole map	Asks before Whole map ⚠	On , Off

- The adventure takes speed and spacing from your **Listen** profile the first time you open it; after that they are its own values. You can also change them from the speed bar, and they apply at once, even while playing.
- **First sentence:** up to the first full stop, including a room name before it. **Room / message:** only the name when you enter a room, otherwise the first sentence of the answer.
- **After playing:** each word appears once it has been played. **Only on tap:** the new answer stays hidden until you tap **Text** or the box.
- **Brief** = BRIEF (the long description on the first visit only), **Superbrief** = SUPERBRIEF (the room name only), **Verbose** = VERBOSE.



Settings: speed & spacing, input



Settings: played in CW, show text, rooms, map

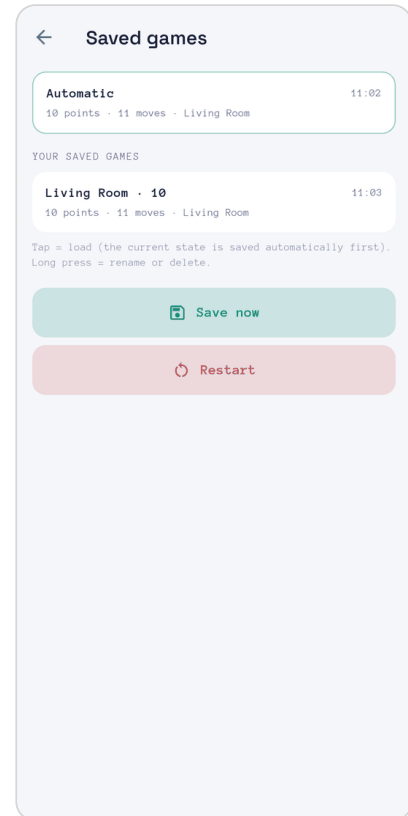
Punctuation in the audio. The screen always shows the text as the game writes it. Only for playing, characters that Morse doesn't have are replaced:

In the text	Played as
' " () [] * # and others	left out
!	▪
;	,
&	AND
Paragraph	double word gap

Saved games.

- **Automatic:** saved after every command and when you leave, one slot per part. Next time you continue exactly there.
- **Your own saves:** as many as you like per part, via : → **Save** or the game command **SAVE**. The name is suggested (room · score).
- **Loading:** via : → **Load**, **Saved games** in the selection, or the game command **RESTORE**. Tap to load; the current state is saved automatically first. Long press to rename or delete.

- **Restart:** : → **Restart** (the app asks first and offers to save before) or the game command **RESTART** (the game asks).
- **Undo** (↶): up to 20 commands while the game screen is open. The original doesn't have this; it mainly helps with typing mistakes.
- **End of the game:** after **QUIT** the app offers to load a saved game, take back the move, or restart.



Saved games of one part

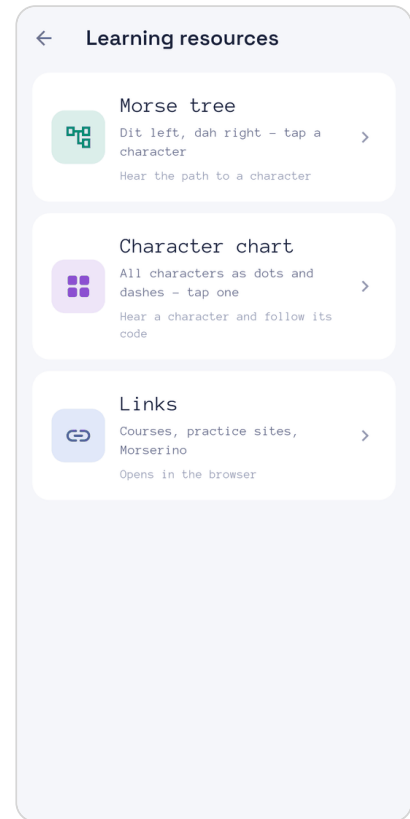
14 Learning resources

Under **Learn** → **Learning resources** you find help for learning CW that is not a training of its own.

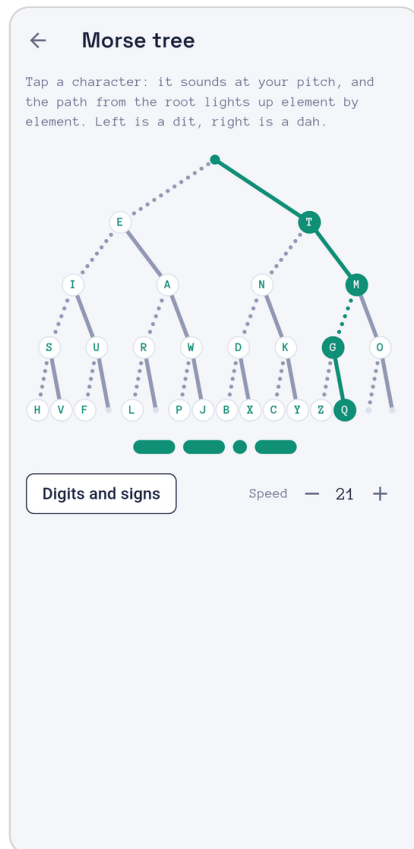
14.1 Morse tree

The Morse tree shows all codes in one picture: from the dot at the top, a **dit goes left** and a **dah goes right**. The lines show it too: a dit is a dotted line, a dah a thick solid one. A character sits where its code ends, so you can read every code by following the path down to it. Letters are on the first four levels. **Digits and signs** adds the fifth level with digits and signs; the tree then scrolls sideways.

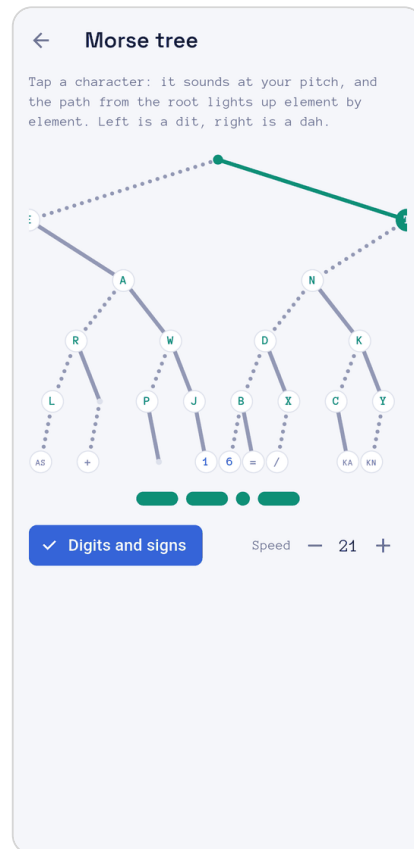
Tap a character to hear it. It sounds at your **pitch** (Settings → General) and at the speed shown below the tree. That speed starts at your **Listen** speed and can be changed here with – and + (10–60 WPM); the change applies only to this screen. While the character sounds, the path from the top lights up element by element: dit or dah is highlighted exactly when you hear it. Below the tree the code is shown once more as dits and dahs.



Learning resources: Morse tree, character chart, links

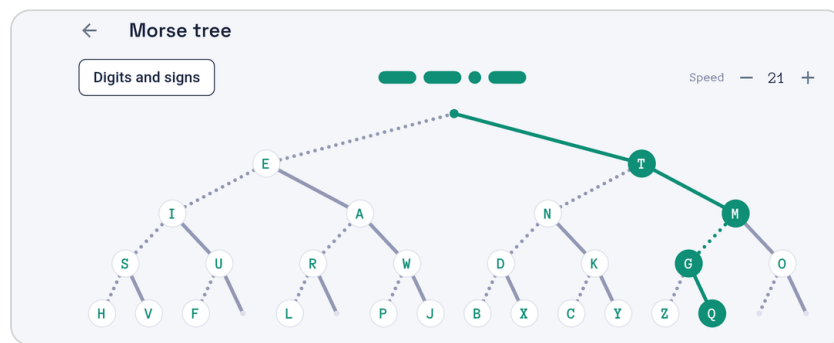


Morse tree: the path to Q lights up

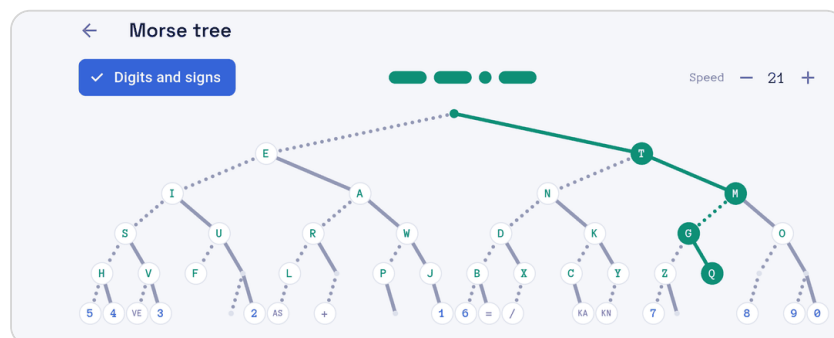


With digits and signs: the tree scrolls sideways

Turn the phone sideways for a bigger tree: the tree screen is the only one in the app that also works in landscape. There the controls sit in one row above the tree, and with **Digits and signs** the whole tree fits on the screen without scrolling. The rest of the app stays in portrait.



Landscape: letters



Landscape with digits and signs: the whole tree fits

The tree is drawn by the app from the same code table it uses for playing and decoding, so it always matches what you hear.

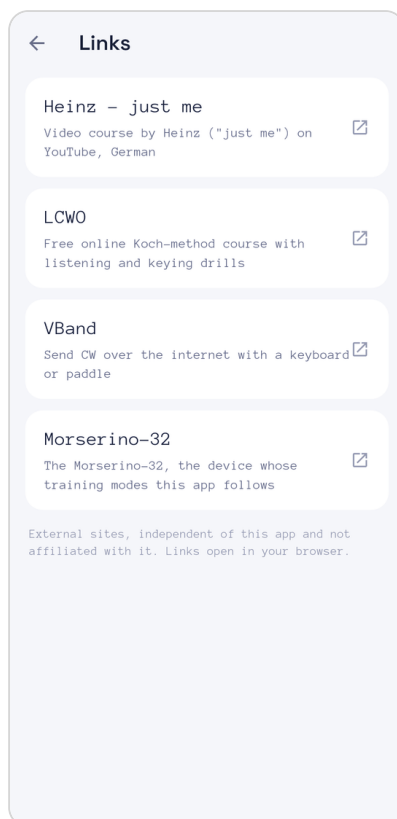
14.2 Character chart

The character chart lists every character of the app with its code, grouped into **Letters**, **Digits**, **Punctuation** and **Prosigns** (SK, KN, KA, AS, VE, BK). Each code is drawn as dots and dashes.

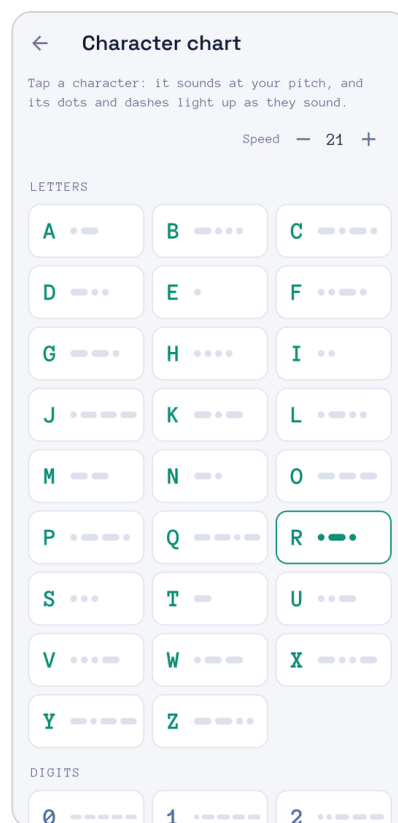
Tap a character: it sounds at your pitch, and its dots and dashes light up as they sound. The speed starts at the value from the Listen training; – and + change it for this view only. Set the pitch under Settings → General.

14.3 Links

A list of external sites for learning CW: the video course by Heinz ("just me") on YouTube, LCWO, VBand and the Morserino-32 homepage. A tap opens the site in your browser. These are independent offers that are not affiliated with this app; the app only links to them and shows nothing of their content.



Links to courses and practice sites



Character chart: a character lights up while it plays

15 Paddle and Morse key

15.1 Touch paddles

Every mode where you key shows two areas at the bottom: **DIT** (left) and **DAH** (right). In the **Straight** keyer mode there is a single **KEY** area that sounds for as long as you press it.

15.2 A real paddle or straight key

A phone has no paddle input. You need a small USB adapter that turns the paddle contacts into **key presses**:

- **vband** (hamradio.solutions/vband) is a widely used, ready-made USB adapter built exactly for this.
- **Homemade**: any small USB HID device that reports the dit and dah contacts as two different keys works. A ready-to-build example is [xiao-vband-adapter](#). It is a Sseed XIAO SAMD21 with a 3.5 mm jack that sends the same keys as the vband adapter. It plugs straight into the phone with a USB-C cable.

Older phones with micro-USB need a USB OTG adapter.

15.2.1 Learning the paddle keys

This tells the app which key your adapter sends for dit and which for dah:

1. Plug in the adapter.
2. Go to **Settings → vband Paddle → Learn paddle keys**.
3. When “Press Dit key ...” appears, press the **dit paddle**. When “Press Dah key ...” appears, press the **dah paddle**.
4. “Saved” confirms it. The detected keys are listed under **Dit** and **Dah**.

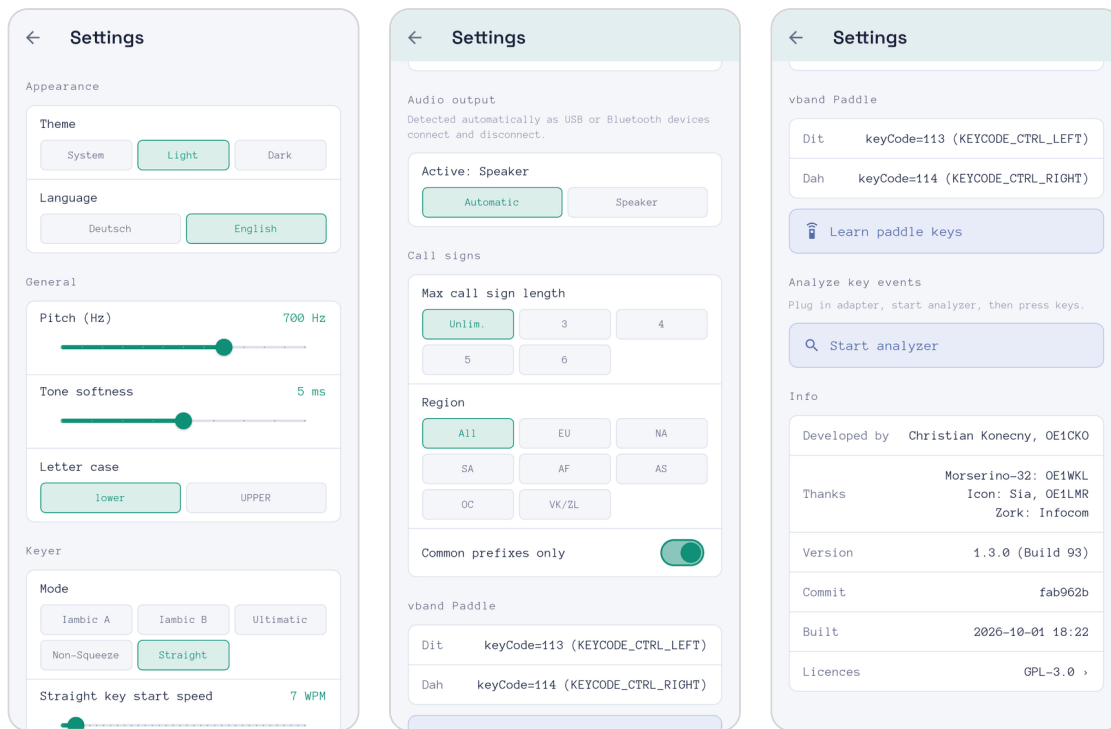
So there's no list of supported adapters. The app learns whatever keys your adapter sends.

15.2.2 Analyze key events

If an adapter doesn't behave as expected, use **Settings → Analyze Key Events**. Press **Start analyzer**, then press the paddles. The app lists every key event it receives, so you can see whether the adapter sends anything and what it sends. **Stop analyzer** ends the display.

16 Settings

Open the global settings with the gear at the top right of the home screen. They apply to the whole app. Anything that concerns only one training is in that training's * sheet.



Settings: Appearance, General, Keyer

Audio output and call signs

vband Paddle, key events, Info

16.1 Appearance

Setting	Meaning	Values
Theme	Light or dark look	System / Light / Dark
Language	App language	Deutsch / English

16.2 General

Setting	Meaning	Values
Pitch (Hz)	Frequency of the sidetone and of the played characters	300–900 Hz in 50 Hz steps (600 Hz)
Tone softness	Rise and fall time of the tone. Larger values sound softer and click less, especially on short dits	1–9 ms (5 ms)
Letter case	Show characters in lower or UPPER case. Display only	lower / UPPER

16.3 Keyer

These settings apply wherever you key.

Setting	Meaning	Values
Mode	How the keyer reads the paddles (see below)	Iambic A / Iambic B / Ultimatic / Non-Squeeze / Straight
CurtisB dit timing	Iambic B and Ultimatic only: from what percentage of a dit a press on the other paddle is already stored	0–100 % in steps of 5 (75 %)
CurtisB dah timing	The same for dahs	0–100 % in steps of 5 (45 %)
Auto character spacing	Enforces a minimum pause between characters so they don't run together. Not available for Straight	Off / 2 / 3 / 4 dits
Straight key start speed	Straight only (replaces Auto character spacing): first estimate for the speed measurement, see Straight key	5–40 WPM (15)

The keyer modes

- **Iambic A:** if you hold both paddles (“squeeze”), dits and dahs alternate. When you let go, the keyer stops after the current element.
- **Iambic B:** like A, but the keyer remembers a press on the other paddle that comes during an element, and adds that element (Curtis B behaviour). The CurtisB settings control from when this applies. 0 % means during the whole element, and 100 % means practically like Iambic A.
- **Ultimatic:** when both paddles are pressed, the **last** one pressed wins and repeats for as long as it is held.
- **Non-Squeeze:** for single-lever paddles or those switching over. Squeezing both paddles produces no alternating sequence.
- **Straight:** a straight key. The tone is on for as long as the key is pressed. With touch, a single **KEY** area appears. With an adapter, the dit contact acts as the key. The speed is measured, see below.

16.3.1 Straight key: automatic speed

With the straight key you don't set a speed. The app measures it from your keying, like the Morserino: from the length of your dits and dahs (a running average) it derives the speed, the dit/dah threshold and the character and word gaps. If you key more slowly, longer pauses are allowed automatically.

- The first estimate is the **Straight key start speed** (Settings → Keyer). If you key much more slowly, set it to your speed, otherwise a slow S can be read as three E at first.

- After about two to four characters the measurement has settled. The very first character can still be misread.
- The measurement restarts every time you enter a screen.
- The **WPM slider** is disabled and moves by itself to your measured speed (Keyer, WiFi Trx, Echo Trainer, QSO Bot and others). It is not saved and does not change the paddle speed.
- WiFi Trx sends the measured speed along. The Echo Trainer statistics store it as your sending speed.

16.4 Audio output

Setting	Meaning	Values
Active	Shows where the sound is going right now	–
Output	Automatic follows whatever is currently plugged in or connected. The other options fix the output. Only outputs that are currently available are offered	Automatic / Speaker / Wired/USB / Bluetooth

16.5 Call signs

These are the settings for random call signs in the **Call signs** content (with All characters).

Setting	Meaning	Values
Max call sign length	Maximum call sign length	Unlim. / 3 / 4 / 5 / 6
Region	Only call signs from this region	All / EU / NA / SA / AF / AS / OC / VK/ZL
Common prefixes only	Only frequently heard prefixes instead of all possible ones	Off / On

Call signs follow a weighted prefix table, as on the Morserino. Frequently heard countries come up more often.

16.6 vband Paddle and Analyze key events

See Learning the paddle keys and Analyze key events.

16.7 Info: version and build

Row	Meaning
Developed by	Christian Konecny, OE1CKO
Thanks	Morserino-32 (OE1WKL), app icon (Sia, OE1LMR), Zork (Infocom)
Version	Version number and build number, for example “1.0.0 (Build 42)”

Row	Meaning
Commit	The exact source code state the app was built from
Built	Date and time of the build
Licences	Tap to open the licence texts (see below)

When you report a problem, please include the version, commit and build time.

Licences: The app is free software under the GNU General Public License v3.0 (or later). It takes algorithms and data (word lists, abbreviations, call sign prefixes, QSO texts) from the Morserino-32 firmware by Willi Kraml, OE1WKL, which is also under the GPL-3.0. The source code is on GitHub. The licence page also shows the MIT License of Zork I–III, the SIL Open Font License of the fonts Anonymous Pro and Space Grotesk, the zlib licence of the SoLoud audio engine, and the licences of the Flutter packages the app uses. Zork is a trademark of its owners; the app is not affiliated with them, nor with Infocom, Activision or Microsoft.

16.8 Koch sequence

You set the Koch sequence in the ***** sheet of **Listen** or **Send** when the **Koch lesson** character set is selected there. It applies to **all** trainings and games, though.

Sequence	Description
M32	The Morserino-32 order (45 characters): <code>m k r s u a p t l o w i . n j e f 0 y v , g 5 / q 9 z h 3 8 b ? 4 2 7 c 1 d 6 x - = + @ :</code>
LCWO	The lcwo.net order
CW Academy	The CW Academy (CWops) order
LICW	The Long Island CW Club order with an entry point (see below)
Custom	Your own order

LICW entry point (0–13): in the LICW course, students join a “carousel” at different points. The entry point rotates the sequence so that it starts there.

Custom: enter the characters in the order you want to learn them. Duplicates are ignored, and the app shows how many characters it found. The default is `esno0tqr5ucd9a18ix1myj7h4gvkfz3b.6/w2p?`, the order of the YouTube Morse course by “Heinz – just me” (playlist).

Prosigns are not part of the Koch sequences in the app.

16.9 Morserino terms

The app gives the Morserino-32 menu items plain names. If you come from the Morserino or read its manual, this table helps:

Morserino menu	In the app
Interchar Spc	Character spacing
InterWord Spc	Word spacing
Random Groups	Group characters
Length Rnd Gr	Group length
Length Words	Max word length
Length Abbrev	Max abbreviation length
Length Calls	Max call sign length
Calls Region	Region (under Call signs)
Max # of Words	Groups per block / Words per block
Stop<Next>Rep	Stop after each group / Stop after each word
Koch Sequence	Koch sequence
Practice Set	Practice set
Boost Practice	Boost practice set
Echo Prompt	Prompt
Echo Repeats	Repeats
Echo Speed Max	Sending speed (max)
Tone Shift	Tone shift
Confrm. Tone	Confirmation tone
AutoChar Spc	Auto character spacing
Output Case	Letter case
Keyer Mode	Mode (under Keyer)
CurtisB DitT% / DahT%	CurtisB dit timing / dah timing

The names of the keyer modes (Iambic A, Ultimatic ...) and of the Koch sequences (M32, LCWO ...) are the same as on the Morserino.

17 What the app does not do (yet)

Some Morserino-32 features are missing from the app, because a phone doesn't have the hardware or Android handles it already. These are the rotary knob and buttons, the display, LoRa, ESP-NOW (and with it the multiplayer parts of the games), iCW/Ext Trx and keying a real transmitter, firmware updates and the WiFi setup page. Instead of the firmware's Practice Stats, the app has its own, more detailed character statistics.

Not done yet, but planned: the games Trailblazer, Fox Hunt, Radio Cave and Fight the Pileup.

18 Troubleshooting

The app doesn't key when I press my paddle. Learn the paddle keys (see Learning the paddle keys). If nothing arrives, check with **Analyze key events** whether the adapter sends anything at all.

Sound comes out of the wrong device. Fix the output under **Settings → Audio output**.

When sending, I hear my tone late. This is almost always Bluetooth headphones. Use wired headphones or the speaker.

The next Koch character doesn't come. Open the  statistics in Listen. The characters without a tick are holding up the unlock. Usually it's a character with too few attempts or recent errors. See When the next Koch character comes.

The pauses never get shorter. That's intended while you work through the Koch sequence (see Pauses and speed). You can shorten them yourself at any time with **Adjust spacing**.

The decoder only writes garbage. Check pitch, bandwidth and threshold (see Decoder settings). Only turn on the monitor tone with headphones.